



Project Report On

**CC-EAM : A Predictive Model for Heart Attack
Detection using Blood Biomarkers**

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CERTIFICATE

*This is to certify that the project report entitled "**CC-EAM : A Predictive Model for Heart Attack Detection using Blood Biomarkers**" is a bonafide record of the work done by **Namitha Mariam John(U2203145)**, **Nandakishore Suresh Nair(U2203146)**, **Nikitha Sarah John (U2203162)**, **Smiya Shaji(U2203195)** submitted to Rajagiri School of Engineering & Technology (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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Abstract

Cardiovascular diseases, particularly myocardial infarction (heart attack), remain one of the leading causes of mortality worldwide. Timely detection and accurate risk prediction are essential to prevent severe complications and reduce death rates. This project presents an AI-Driven Heart Attack Detection and Prediction System designed to analyze clinically significant biomarkers such as Age, Troponin, and CK-MB to determine the likelihood of myocardial infarction.

The proposed system utilizes a Clinically Constrained Explainable Additive Model (CC-EAM), which ensures both high predictive performance and medical reliability by incorporating clinically meaningful constraints. Unlike conventional black-box machine learning models, the system provides a quantified risk probability score along with transparent, feature-wise explanations, clearly showing how each biomarker contributes to the final prediction.

The model is optimized for high recall and strong ROC-AUC performance to minimize false negatives in critical cases. By combining predictive analytics with explainable artificial intelligence, the system aims to support clinicians in early diagnosis, assist in risk stratification, and improve decision-making in cardiovascular care.

Contents

Acknowledgements	i
Abstract	ii
List of Abbreviations	vi
List of Figures	vii
List of Tables	viii
1 Introduction	1
1.1 Background	1
1.2 Problem Definition	2
1.3 Scope and Motivation	2
1.4 Objectives	3
1.5 Challenges	3
1.6 Assumptions	4
1.7 Societal / Industrial Relevance	5
1.8 Organization of the Report	6
1.9 Summary	7
2 Literature Survey	8
2.1 Review of Key Research Works	8
2.1.1 Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI	8
2.1.2 A Biomarker Discovery of Acute Myocardial Infarction Using Fea- ture Selection and Machine Learning	10
2.1.3 Machine Learning for Early Prediction of Acute Myocardial Infarc- tion or Death in Acute Chest Pain Patients	11

2.1.4	Prognostic Utility of a Multi-Biomarker Panel in Patients with Sus- pected Myocardial Infarction	13
2.2	Comparative Analysis	14
2.3	Summary and Gaps Identified	16
3	Proposed Method	17
3.1	Overview	17
3.2	Overall System Workflow	17
3.3	System Architecture	18
3.4	Module-wise Architecture and Description	19
3.4.1	Input Preprocessing Module	19
3.4.2	Risk Prediction Module	20
3.4.3	Clinically Constrained Explainable Additive Model (CC-EAM) . . .	20
3.4.4	Explainable AI Module	21
3.4.5	Medical Report Generation Module	21
3.4.6	Medical Literature Recommendation Module	22
3.5	Summary	23
4	System Design	24
4.1	Overall System Architecture	24
4.1.1	Database Layer	24
4.1.2	Backend (Model & Logic Layer)	24
4.1.3	Explainable AI Layer	25
4.1.4	Frontend (User Interface)	25
4.2	Design Diagrams	27
4.2.1	Use Case Diagram	27
4.2.2	Sequence Diagram	28
4.3	Algorithms Used	29
4.3.1	Algorithm 1: Logistic Regression (LR)	29
4.3.2	Algorithm 2: Support Vector Machine (SVM)	30
4.3.3	Algorithm 3: Random Forest (RF)	30
4.3.4	Algorithm 4: XGBoost (Extreme Gradient Boosting)	31
4.3.5	Algorithm 5: Explainable Boosting Machine (EBM)	31

4.3.6	Algorithm 6: CCEAM (Clinically Constrained Explainable Additive Model)	32
4.4	Software and Hardware Requirements	33
4.4.1	Software Requirements	33
4.4.2	Hardware Requirements	33
4.5	Dataset Identified	33
4.5.1	Dataset	34
4.6	Project Timeline	34
4.7	Module Division and Work Responsibilities	35
4.8	UI Design	37
4.9	Expected Outcomes	37
4.10	Summary of Chapter	38
5	Results and Discussions	39
5.1	Overview	39
5.2	Results	40
5.3	Output	42
5.4	Discussion	45
5.5	Summary	46
6	Conclusion	47
6.1	Conclusion	47
6.2	Future Scope	48
6.3	Summary	48
	References	48
	Appendix A: Presentation	51
	Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes	77
	Appendix C: CO-PO-PSO Mapping	81

List of Abbreviations

MI	– Myocardial Infarction
CC-EAM	– Clinically Constrained Explainable Additive Model
Apo A-I	– Apolipoprotein A-I
AUPRC	– Area Under the Precision-Recall Curve
CK-MB	– Creatine Kinase-Myocardial Band
CRP	– C-Reactive Protein
ECG	– Electrocardiogram
ED	– Emergency Department
GRACE	– Global Registry of Acute Coronary Events
HDL	– High-Density Lipoprotein
HR	– Hazard Ratio
hs-cTn	– High-Sensitivity Cardiac Troponin
KIM-1	– Kidney Injury Molecule-1
LASSO	– Least Absolute Shrinkage and Selection Operator
LDL	– Low-Density Lipoprotein
LIME	– Local Interpretable Model-agnostic Explanations
MACE	– Major Adverse Cardiovascular Events
NPV	– Negative Predictive Value
NRI	– Net Reclassification Improvement
NT-proBNP	– N-terminal pro B-type Natriuretic Peptide
PCC	– Pearson Correlation Coefficient
SHAP	– SHapley Additive exPlanations
SMOTE	– Synthetic Minority Over-sampling Technique
TIMI	– Thrombolysis in Myocardial Infarction

List of Figures

3.1	Overall System Architecture	19
3.2	Risk Prediction Module	20
3.3	Clinically Constrained Explainable Additive Model	21
3.4	Report Generation Module	22
3.5	Architecture of the Medical Literature Recommendation Module	23
4.1	Use Case Diagram	27
4.2	Sequence Diagram	28
4.3	Project Timeline (Gantt Chart)	34
4.4	UI Design - Overview Screen	37
5.1	Model Performance Comparison 1	40
5.2	Model Performance Comparison 2	40
5.3	ROC Curves for all Models	41
5.4	CCEAM model evaluation	41
5.5	Risk Assessment Dashboard	42
5.6	Recommended Medical Literature	42
5.7	CCEAM Troponin Risk (High)	43
5.8	CC-EAM Troponin Risk (Low)	43
5.9	MI Prediction Report	44
5.10	Nearby Hospital Recommendations	44
5.11	Medical Literature Search	45

List of Tables

2.1	Comparison of key studies on biomarker-based machine learning for heart attack prediction	15
4.1	Work Division Chart	35

Chapter 1

Introduction

1.1 Background

Cardiovascular diseases remain one of the most pressing public health challenges of the 21st century, accounting for millions of deaths globally each year. Among these, myocardial infarction, commonly known as a heart attack, stands out as a leading cause of sudden mortality, often striking without adequate warning. Despite significant advances in modern medicine, a considerable number of cardiac events go undetected until they reach a critical or irreversible stage. The window between early onset and severe complications is narrow, making timely and accurate detection not merely beneficial, but lifesaving.

Traditional diagnostic approaches for identifying MI rely heavily on clinical judgment, ECGs, and laboratory tests interpreted by trained cardiologists. While effective in controlled hospital environments, these methods are often inaccessible in under-resourced settings, time intensive, and high-pressure conditions. Furthermore, conventional machine learning models, though increasingly applied in medical diagnostics, frequently operate as black-box systems offering predictions without interpretable reasoning, which limits their acceptance and trustworthiness in clinical practice.

Recognizing these challenges, the AI-Driven Heart Attack Detection and Prediction System emerges as a modern, clinically grounded solution designed to bridge the gap between predictive analytics and real-world medical utility. By analyzing key biomarkers such as Age, Troponin, and CK-MB, indicators with well established clinical significance in cardiac events, the system delivers a quantified risk probability score that reflects the likelihood of myocardial infarction with high statistical reliability.

At the core of the system lies the Clinically Constrained Explainable Additive Model (CC-EAM), which distinguishes itself from conventional approaches by embedding medically meaningful constraints directly into the model architecture. This ensures that

predictions not only achieve high performance metrics , particularly in recall and ROC-AUC, but also align with established clinical knowledge, thereby reducing the risk of medically implausible outputs. Crucially, the model offers transparent, feature wise explanations that clearly articulate how each biomarker influences the final risk assessment, empowering clinicians to understand, verify, and trust the system’s reasoning.

Overall, the AI-Driven Heart Attack Detection and Prediction System addresses the urgent need for intelligent, explainable, and medically reliable tools in cardiovascular care. By combining the power of artificial intelligence with the rigor of clinical constraints, it aims to support early diagnosis, facilitate risk stratification, and ultimately improve patient outcomes in both clinical and resource-limited healthcare settings.

1.2 Problem Definition

Problem: Cardiovascular diseases, particularly myocardial infarction, are a leading cause of global mortality. Existing diagnostic methods are time-intensive, heavily reliant on specialist availability, and conventional ML models lack transparency, making them difficult to trust and adopt in clinical settings.

Solution: To develop an AI-driven system that analyzes clinically significant biomarkers like age, troponin, and CK-MB, to predict the risk of myocardial infarction with high recall, while providing transparent, feature wise explanations to support clinical decision-making.

The aim of the project is to create an AI-Driven Heart Attack Detection and Prediction System that enhances cardiovascular care through a Clinically Constrained Explainable Additive Model (CC-EAM), combining accurate risk prediction with interpretable, medically reliable insights to assist clinicians in early diagnosis and risk stratification.

1.3 Scope and Motivation

Scope: The scope of this project encompasses the design, development, and testing of the AI-Driven Heart Attack Detection and Prediction System. This includes building a clinically grounded platform that integrates biomarker based risk prediction using a Clinically Constrained Explainable Additive Model (CC-EAM), capable of analyzing key indicators such as age, troponin, and CK-MB to assess the likelihood of MI. The system will feature

a quantified risk probability score along with transparent, feature-wise explanations that clearly illustrate how each biomarker contributes to the final prediction.

Motivation: The development of the AI-Driven Heart Attack Detection and Prediction System is motivated by the critical global need to reduce mortality from undetected or late diagnosed MI. Despite medical advancements, many cardiac events are identified only after irreversible damage has occurred. This is largely due to specialist shortages, the time sensitive nature of diagnosis, and the opacity of existing AI tools. Conventional machine learning models are often distrusted in healthcare environments because of their "black box" nature. By combining predictive analytics with Explainable AI (XAI) and clinically meaningful constraints, this system bridges the gap between computational power and clinical trust. Ultimately, it empowers healthcare professionals with a reliable, transparent decision-support tool to enable early diagnosis, enhance risk stratification, and improve cardiovascular patient outcomes.

1.4 Objectives

1. To develop an AI-based system for early prediction of myocardial infarction (heart attack) using clinical biomarker data.
2. To implement a Clinically Constrained Explainable Additive Model (CC-EAM) for accurate and clinically consistent risk prediction.
3. To generate a probability-based MI Risk Score instead of only binary classification.
4. To integrate Explainable AI (XAI) for interpreting how each biomarker contributes to the prediction
5. To create a user-friendly interface that is accessible across various devices that clearly displays the risk score, risk category, and explanation..

1.5 Challenges

- Ensuring the accuracy and generalizability of the risk prediction model, particularly across diverse patient demographics, varying medical histories, and different labora-

tory measurement standards for biomarkers (such as Troponin and CK-MB), poses a significant challenge requiring robust model tuning and validation.

- Maintaining patient privacy and data security while handling highly sensitive medical records and clinical biomarkers necessitates the careful implementation of encryption, data anonymization, and strict access controls to comply with healthcare regulations (e.g., HIPAA/GDPR).
- Achieving reliable and medically meaningful explainability from the CC-EAM involves challenges related to embedding accurate clinical domain constraints, handling complex feature interactions, and ensuring the output is intuitively trusted by clinicians, demanding rigorous optimization and expert validation.

1.6 Assumptions

1. The biomarker values (e.g., Troponin, CK-MB) input into the system are reliable, obtained from accurate laboratory tests, and entered correctly by the user or clinician.
2. The user interface will be accessed by healthcare professionals or informed patients who have a basic understanding of medical terms and can effectively interact with the web dashboard.
3. The historical patient data used to train the Clinically Constrained Explainable Additive Model (CC-EAM) is representative of real-world patient cases and reflects similar clinical characteristics as the target deployment population.
4. A stable internet connection will be available to ensure seamless communication between the frontend and the backend ML models, API calls to PubMed for document retrieval, and the nearest hospital locator.
5. The system is intended to function as a supplementary clinical decision-support tool rather than a standalone diagnostic replacement, with final medical decisions remaining in the hands of qualified healthcare professionals.

6. External clinical factors or diagnostic tests not explicitly included in the model’s dataset (e.g., ECG findings, imaging reports) are assumed not to drastically contradict the risk interpretation provided by the blood biomarkers.

1.7 Societal / Industrial Relevance

The AI-Driven Heart Attack Detection and Prediction System holds significant societal and industrial relevance by addressing the critical global health challenge of cardiovascular disease mortality. In a society where heart attacks frequently occur without adequate warning or are diagnosed too late, this system provides a lifesaving tool for the early detection of myocardial infarction. It is particularly beneficial in rural or under-resourced areas where specialist cardiologists are scarce, offering patients and local healthcare providers immediate, reliable risk assessments and directing them to the nearest cardiac care centers during emergencies.

From an industrial and healthcare perspective, hospitals, emergency departments, and diagnostic laboratories can integrate this system to accelerate triage and improve clinical decision-making. By utilizing a Clinically Constrained Explainable Additive Model (CC-EAM), medical institutions can trust the AI’s transparent reasoning, bridging the gap between advanced predictive analytics and practical clinical application. This leads to more efficient patient management, optimized allocation of emergency medical resources, and a reduction in the severe physical and financial burdens associated with late-stage cardiac events. Its implementation in clinical workflows demonstrates its immense potential to improve both societal health outcomes and the operational efficiency of the healthcare industry.

1.8 Organization of the Report

This report presents the framework for developing an AI-powered system for the detection and prediction of heart attacks using explainable machine learning techniques. It begins by addressing the fundamental clinical challenges of accurately identifying myocardial infarction (MI), noting that symptoms such as chest pain often originate from diverse causes, leading to diagnostic uncertainty and unnecessary hospital evaluations. The discussion progresses to establish the need for a data-driven diagnostic approach that leverages blood biomarkers to achieve early and reliable risk prediction.

The report is systematically structured to provide a clear and logical flow from conceptual understanding to technical implementation.

Chapter 1 introduces the project background, outlines the problem definition, scope, objectives, risks, and challenges, and highlights the societal and industrial relevance of the proposed system.

Chapter 2 reviews previous studies and existing diagnostic methodologies, comparing conventional approaches such as ECG and cardiac risk scores with modern machine learning models. It emphasizes the role of biomarkers such as Troponin, CK-MB, CRP, D-dimer, and Homocysteine, and identifies research gaps motivating the proposed AI-integrated system.

Chapter 3 details the system design and architecture, describing how the database layer, backend model and logic layer, and frontend clinician interface are interconnected. It discusses the design components, including data preprocessing, feature engineering, model training and selection and Explainable AI modules. The chapter also includes dataset details, software and hardware requirements, module responsibilities, UI design, and a project timeline outlining the implementation plan.

Chapter 4 Presents the proposed methods for this system.

Finally, Chapter 5 concludes the report by summarizing the key project achievements, expected deliverables, and future directions. It emphasizes how the proposed system integrates biomarker-based analytics with interpretable AI models to improve early heart attack detection, clinical decision-making, and overall cardiovascular care and suggests possible enhancements and extensions that can be incorporated in future iterations. The report is thus organized to progress logically from theoretical foundation to practical

system design.

1.9 Summary

In today’s fast-paced world, cardiovascular disease, particularly myocardial infarction, has become a leading cause of global mortality. The inability to predict and diagnose a heart attack accurately in its early stages often leads to irreversible cardiac damage. To address this, the AI-Driven Heart Attack Detection and Prediction System leverages a Clinically Constrained Explainable Additive Model (CC-EAM) to analyze clinically significant blood biomarkers, such as Troponin and CK-MB. The system computes a precise, probability-based risk score while providing transparent, human-readable explanations. It delivers rapid risk stratification and recommends nearby hospitals and relevant medical literature, acting as a crucial decision-support tool.

The project aims to develop a robust, integrated platform that bridges the gap between complex artificial intelligence and clinical trust. Objectives include replacing traditional binary classification with probabilistic scores, embedding clinical constraints to ensure medically valid outputs, providing interpretable explanations for every prediction, and offering actionable insights through automated diagnostic reports.

Challenges include ensuring the model’s accuracy across diverse patient demographics, maintaining strict data privacy for sensitive health metrics, handling imbalanced datasets, and achieving reliable generalizability under real-world clinical conditions.

Assumptions include user access to accurate laboratory biomarker test results, internet connectivity for seamless API integration (like PubMed and hospital search), and the understanding that the platform functions as an assistive tool rather than a standalone diagnostic replacement.

The system holds societal and industrial relevance by addressing the urgent need for early cardiac intervention, supporting healthcare providers in under-resourced areas, and improving hospital triage efficiency. The report’s organization includes sections on the introduction, literature survey, system architecture and design, and module descriptions.

Chapter 2

Literature Survey

This project reviews recent advancements in heart attack detection and prognosis with a focus on studies that integrate biomarker analysis and machine learning techniques. The discussion draws primarily from benchmark studies such as “Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI” [3] and “Machine Learning for Early Prediction of Acute Myocardial Infarction or Death in Acute Chest Pain Patients” [2], along with other key research on biomarker discovery and prognostic modeling. These works collectively demonstrate the feasibility of using blood biomarkers (e.g., Troponin, CK-MB, CRP, NT-proBNP, Apo A-I, KIM-1) in predictive frameworks enhanced by Explainable AI (XAI) methods such as SHAP and LIME. The review emphasizes each study’s contributions, methodologies, and findings, while identifying their strengths, limitations, and existing research gaps that motivate the design of the proposed AI-powered, interpretable heart attack detection system.

2.1 Review of Key Research Works

2.1.1 Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI

The paper titled “Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI” [3] presents a comprehensive approach to improving heart attack prediction by integrating Artificial Intelligence (AI) models with Explainable AI (XAI) techniques to enhance both accuracy and interpretability. The study addresses the critical need for early detection of heart attacks, where timely and accurate diagnosis can significantly reduce morbidity and mortality rates. Traditional diagnostic tools such as ECGs and stress tests, though valuable, often fail to capture subtle early-stage indicators of cardiovascular disease. Hence, the authors explore the

use of AI-based systems that analyze multiparametric cardiac data to identify underlying patterns not easily visible through conventional methods.

The researchers used the UCI Heart Disease Dataset, which contains 303 patient records encompassing 14 validated clinical features such as age, sex, resting blood pressure, cholesterol levels, fasting blood sugar, ECG readings, exercise-induced angina, ST depression (oldpeak), and Thallium test results (thal). A systematic preprocessing pipeline was implemented, which included handling missing values, normalizing features using Min-Max scaling, and addressing class imbalance through the Synthetic Minority Over-sampling Technique (SMOTE). Categorical attributes were also encoded to make the dataset compatible with machine learning algorithms.

A comparative study of several machine learning algorithms—including Logistic Regression, Naïve Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), and ensemble models like AdaBoost and Bagging—was conducted. Among all tested models, the Artificial Neural Network (ANN) demonstrated the best predictive capability, achieving 96.1% accuracy, 95.7% recall, and an F1-score of 95.7% during 10-fold cross-validation. The ANN architecture consisted of one input layer with 12 neurons, one hidden layer with 8 neurons using ReLU activation, and one output layer with a Sigmoid function. Optimization was performed using the Adam optimizer with binary cross-entropy as the loss function. Hyperparameter tuning was achieved through grid search to maximize performance.

The study also emphasized model explainability through the incorporation of XAI techniques—SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) [7]. SHAP provided global feature importance and interaction effects, highlighting the most influential predictors such as the number of major vessels (ca), chest pain type (cp), Thallium test result (thal), ST depression (oldpeak), and maximum heart rate achieved (thalach). LIME offered local interpretability by generating per-patient explanations, indicating which specific features contributed to individual predictions. For example, patients with high oldpeak, abnormal thal values, and low thalach were predicted as high-risk, aligning with medical understanding of heart disease pathology.

In conclusion, this study demonstrates that combining Artificial Neural Networks with Explainable AI significantly improves both the diagnostic accuracy and interpretability

of heart attack prediction systems. By providing a clear rationale for each prediction, the model builds confidence among healthcare professionals, facilitating better clinical decision-making. Moreover, the research highlights the potential of such AI-driven systems to be extended and adapted for other cardiovascular datasets, paving the way for future deployment in real-world medical environments as reliable decision-support tools.

2.1.2 A Biomarker Discovery of Acute Myocardial Infarction Using Feature Selection and Machine Learning

Cardiovascular diseases continue to be a major global health concern, with acute myocardial infarction (AMI) being one of the most critical and life-threatening conditions. Rapid and accurate diagnosis of AMI is essential for effective patient management, as diagnostic delays can significantly increase morbidity and mortality [5]. Traditional diagnostic tools such as electrocardiography (ECG), enzyme assays, and imaging provide valuable information but often lack the predictive precision required for early intervention. [11] To overcome these limitations, this study proposed an artificial intelligence-driven approach that integrates feature selection and machine learning (ML) for biomarker discovery and AMI detection.

Input and Dataset: The research utilized a clinical dataset comprising both demographic and biochemical information from patients with suspected or confirmed AMI. The features included age, gender, blood pressure, cholesterol levels, triglycerides, C-reactive protein (CRP), troponin, CK-MB, and other key biomarkers. Each sample was labeled as either positive or negative for myocardial infarction based on confirmed clinical diagnoses. This rich dataset allowed the identification of significant biomarkers correlated with AMI risk.

Methodology: The data underwent preprocessing to ensure consistency and accuracy. Missing values were imputed, outliers were treated, and normalization was performed to standardize the feature scales. The study employed several feature selection methods—such as Pearson Correlation Coefficient (PCC), Recursive Feature Elimination (RFE), and Random Forest Importance (RFI)—to extract the most relevant biomarkers contributing to AMI prediction. After feature selection, multiple ML algorithms were trained and tested, including Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), k-Nearest Neighbor (kNN), and Artificial

Neural Network (ANN). Each model was optimized using cross-validation techniques to avoid overfitting and ensure robust generalization.

Evaluation Metrics: Model performance was evaluated using standard metrics such as accuracy, precision, recall, F1-score, and the Area Under the Precision-Recall Curve (AUPRC). The confusion matrix was analyzed to assess false positives and false negatives, providing deeper insights into the diagnostic capability of each classifier.

Results and Output: The study found that the Random Forest and Support Vector Machine algorithms outperformed the other models, achieving the highest accuracy and stability in classifying AMI cases. The RFI-based 13-feature model (RFI-13) achieved optimal performance, indicating that a limited number of well-chosen biomarkers can yield high diagnostic reliability. Simpler algorithms like Naïve Bayes and Decision Tree showed relatively lower performance due to their limited handling of complex nonlinear patterns present in the dataset. The results confirmed that ML-based biomarker discovery significantly enhances diagnostic precision compared to traditional techniques, providing faster, data-driven, and interpretable insights for early detection of myocardial infarction.

Conclusion: This work highlights the clinical potential of integrating ML and feature selection in cardiac diagnostics. By identifying the most influential biomarkers and optimizing predictive models, the study establishes a strong foundation for developing intelligent decision-support systems capable of assisting clinicians in early and accurate AMI diagnosis. It demonstrates how combining advanced analytics with medical knowledge can bridge the gap between raw data and actionable clinical insights.

2.1.3 Machine Learning for Early Prediction of Acute Myocardial Infarction or Death in Acute Chest Pain Patients

Acute myocardial infarction (AMI) remains one of the most critical and time-sensitive cardiovascular emergencies, where rapid and accurate diagnosis is essential to improve patient outcomes. Traditional diagnostic approaches such as ECG interpretation and clinical risk scores often fall short in identifying patients at minimal risk for AMI or short-term mortality [8]. To address these limitations, de Capretz *et al.* [2] proposed a biomarker-driven machine learning (ML) framework for early risk stratification of patients presenting with acute chest pain in the emergency department (ED).

Input and Dataset: The study utilized data from 9,519 consecutive ED patients

presenting with chest pain across multiple clinical centers. Each patient record included demographic characteristics and key biochemical markers measured at presentation, such as high-sensitivity cardiac troponin T (hs-cTnT), glucose, creatinine, and hemoglobin levels. These biomarkers were selected due to their established relevance to cardiac and metabolic health [11]. All cases were annotated for 30-day outcomes — including acute myocardial infarction or death — enabling a robust supervised ML training and validation process.

Methodology: Data preprocessing involved handling missing values, normalization, and scaling to ensure feature consistency. Two predictive models were developed: Logistic Regression (LR) and an Artificial Neural Network (ANN). The LR model offered interpretability and provided insights into the statistical significance of each variable, while the ANN captured complex nonlinear relationships between biomarkers. Fivefold cross-validation was performed to prevent overfitting, and both models were calibrated to maximize sensitivity and negative predictive value (NPV) — critical factors in safely ruling out AMI. The ML models were compared to conventional diagnostic assessments such as troponin-only thresholds and ECG-based evaluations.

Evaluation Metrics: Performance evaluation focused on clinically meaningful indicators, primarily sensitivity and NPV, to ensure diagnostic safety. Secondary metrics included specificity, accuracy, and the area under the receiver operating characteristic curve (AUC). The LR and ANN models were benchmarked against standard care pathways to assess improvement in early triage precision and reduction of unnecessary admissions.

Results and Output: Both the LR and ANN models achieved exceptionally high sensitivity ($>99\%$) and NPV ($>99.5\%$), ensuring minimal false negatives during early AMI detection. The ANN model demonstrated slightly superior discrimination and generalization performance, highlighting its ability to capture complex feature interactions. Importantly, both models outperformed traditional risk assessment tools, underscoring the diagnostic value of integrating biochemical and demographic data for real-time decision support in the ED.

Conclusion: This multicenter retrospective study demonstrates that ML-driven models leveraging routine biomarker data can safely and effectively stratify patients presenting with acute chest pain. By achieving high sensitivity and NPV, the approach supports early discharge decisions without compromising patient safety. These findings highlight

the growing clinical potential of explainable AI in cardiovascular risk prediction and emergency care management.

2.1.4 Prognostic Utility of a Multi-Biomarker Panel in Patients with Suspected Myocardial Infarction

Cardiovascular diseases remain a global health crisis, with acute myocardial infarction (AMI) posing one of the greatest threats due to its high mortality and morbidity. Accurate risk assessment and early identification of patients likely to develop major adverse cardiovascular events (MACE) are vital for effective management. Traditional diagnostic approaches, such as ECG and GRACE scores though valuable, are insufficient for precise prognostication [4]. This study addresses this limitation by implementing a machine learning (ML)-based, multi-biomarker model to predict 1-year cardiovascular outcomes in patients with suspected MI.

Input and Dataset: The research utilized data from 748 consecutive patients enrolled in the Biomarkers in Acute Cardiac Care (BACC) study, all presenting with symptoms indicative of MI. The dataset included 29 biomarkers capturing diverse physiological pathways — such as cardiac stress, inflammation, lipid metabolism, and renal injury — along with demographic and clinical variables (age, sex, blood pressure, diabetes, smoking status, etc.). Each case was adjudicated according to the Fourth Universal Definition of MI, and patient outcomes were tracked for 1 year to record the occurrence of major adverse cardiovascular events (MACE).

Methodology: The study implemented a machine learning–assisted feature selection approach using the Least Absolute Shrinkage and Selection Operator (LASSO) method to identify the most prognostically relevant biomarkers.

Biomarker data were preprocessed through log transformation to normalize distributions. Cox regression analyses (both crude and adjusted for age/sex) were applied to determine hazard ratios (HRs) for each biomarker.

LASSO regression with fivefold cross-validation identified the optimal combination of biomarkers for outcome prediction.

Model performance was benchmarked against standard clinical risk models and the Global Registry of Acute Coronary Events (GRACE) score. Evaluation metrics included the C index, Akaike’s information criterion (AIC), and Net Reclassification Improvement (NRI),

with Kaplan-Meier analyzes validating prediction times per event.

Evaluation Metrics: Predictive performance was measured using discrimination (C-index), calibration, and reclassification improvements compared to established models. Internal cross-validation ensured model generalization and prevented overfitting. Improvement in predictive accuracy was quantified as the C-index and NRI (category-free).

Results and Output: From the 29 biomarkers analyzed, three emerged as the most powerful independent predictors of 1-year MACE:

- NT-proBNP (HR per SD: 1.24) — marker of myocardial stress
- Apolipoprotein A-I (Apo A-I) (HR per SD: 0.98) — inverse indicator of cardiovascular risk
- Kidney Injury Molecule-1 (KIM-1) (HR per SD: 1.06) — marker of renal and vascular dysfunction

The LASSO-selected model (NT-proBNP + Apo A-I + KIM-1 + age + sex) achieved a corrected C-index of 0.63, outperforming the standalone clinical model (0.617) and GRACE score (0.629). When these biomarkers were integrated with the clinical risk model, the C-index increased by 0.047, with an NRI of 0.41, showing significant improvement in correctly reclassifying low-risk (non-MACE) patients.

Conclusion: This study demonstrates that combining machine learning with multi-biomarker profiling can enhance risk prediction in patients with suspected myocardial infarction. The integration of NT-proBNP, Apo A-I, and KIM-1 into standard clinical models markedly improved prognostic accuracy and patient stratification. The approach highlights how AI-driven biomarker selection can move cardiovascular diagnostics beyond traditional metrics, enabling more personalized, data-informed clinical decisions for early detection and management of high-risk patients.

2.2 Comparative Analysis

The reviewed studies demonstrate that integrating machine learning with biomarker-based data significantly improves the accuracy and interpretability of heart attack prediction compared to traditional models. A comparative summary of the methods, biomarkers, and outcomes is presented in Table 2.1.

Table 2.1: Comparison of key studies on biomarker-based machine learning for heart attack prediction

Paper	Focus Area	Key Biomarkers	ML Method/Approach	Key Findings (Biomarkers)
Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI	Heart attack prediction using Explainable AI	LDL, HDL, ECG, Demographic Data	Logistic Regression, KNN, SVM, Random Forest, ANN, SHAP, LIME	ANN achieved highest accuracy; XAI improved interpretability
A Biomarker Discovery of Acute Myocardial Infarction Using Feature Selection and Machine Learning	Biomarker discovery for AMI	Multiple biomarkers selected using PCA, GA, RFE	Logistic Regression, Decision Tree, RF, GBM, SVM, ANN	Feature selection methods improved AMI prediction accuracy
Machine Learning for Early Prediction of Acute Myocardial Infarction or Death in Acute Chest Pain Patients	Early AMI prediction in emergency patients	hs-cTnT, Glucose, Creatinine, Hemoglobin	Logistic Regression, ANN	Achieved >99% sensitivity and NPV; validated biomarker-only prediction
Prognostic Utility of a Multi-Biomarker Panel in Patients with Suspected Myocardial Infarction	Prognostic utility of multi-biomarker panels	NT-proBNP, Apo A-I, KIM-1	LASSO Regression	Integration with GRACE score improved long-term MACE prediction

2.3 Summary and Gaps Identified

The reviewed literature consistently demonstrates that combining multiple biomarkers with advanced machine learning techniques produces more accurate and personalized cardiovascular predictions than traditional, single-biomarker models. Studies show that algorithms such as Random Forest, XGBoost, LASSO, and ANN effectively capture complex relationships within clinical data, while Explainable AI tools enhance interpretability and trust among clinicians.

However, several research gaps remain:

- Dataset limitations: Many studies rely on small or heterogeneous datasets, reducing the generalizability of findings.
- Real-time deployment challenges: Computational complexity and the need for continuous biomarker monitoring limit clinical scalability.
- Interpretability and trust: Despite progress in XAI, ensuring full clinical transparency remains an ongoing challenge.
- Broader applicability: Most models focus solely on myocardial infarction, underscoring the need for frameworks capable of predicting multiple cardiovascular outcomes.

Addressing these gaps is crucial for developing robust, explainable, and clinically viable AI systems—an objective that directly motivates the design of the proposed Heart-Attack Detection and Early Warning System using Explainable AI.

Chapter 3

Proposed Method

3.1 Overview

The proposed system presents an AI-driven framework for the early detection and prediction of myocardial infarction (heart attack) using clinically significant blood biomarkers. The system focuses on analyzing key indicators such as Age, Troponin, and CK-MB to estimate the probability of a cardiac event.

Unlike traditional diagnostic methods that rely primarily on electrocardiogram (ECG) readings or fixed clinical scoring systems, the proposed approach integrates machine learning models with explainable artificial intelligence techniques to provide both accurate predictions and transparent explanations.

At the core of the system lies the Clinically Constrained Explainable Additive Model (CC-EAM), which ensures that predictions remain consistent with medical knowledge while still benefiting from the predictive power of machine learning. The system processes patient biomarker inputs, performs necessary preprocessing, applies trained machine learning models for risk prediction, and generates explainable outputs along with actionable recommendations such as nearby hospital locations and relevant medical literature.

3.2 Overall System Workflow

The overall workflow of the proposed system consists of the following stages:

- Patient biomarker data such as Age, gender, Troponin, and CK-MB are collected through the user interface.
- The input data undergo preprocessing steps including normalization, scaling, and handling of missing values to ensure data consistency.

- The cleaned data is passed to the trained Clinically Constrained Explainable Additive Model (CC-EAM) and supporting machine learning models.
- Multiple prediction pathways including static models and temporal biomarker analysis are executed.
- A probability-based myocardial infarction risk score is computed.
- Based on the computed probability, the patient is classified into risk categories such as Low Risk or High Risk.
- If the risk level is high, the system generates alerts and suggests nearby hospitals for immediate medical attention.
- Explainable AI techniques analyze feature contributions and generate interpretable explanations.
- A structured medical report containing prediction results and explanations is generated and made available for download.
- Relevant medical literature and research articles are retrieved automatically to support clinical decision-making.

3.3 System Architecture

The architecture of the proposed system consists of multiple interconnected components including data input, preprocessing, prediction engine, explainable AI module, report generation system, and medical knowledge retrieval module.

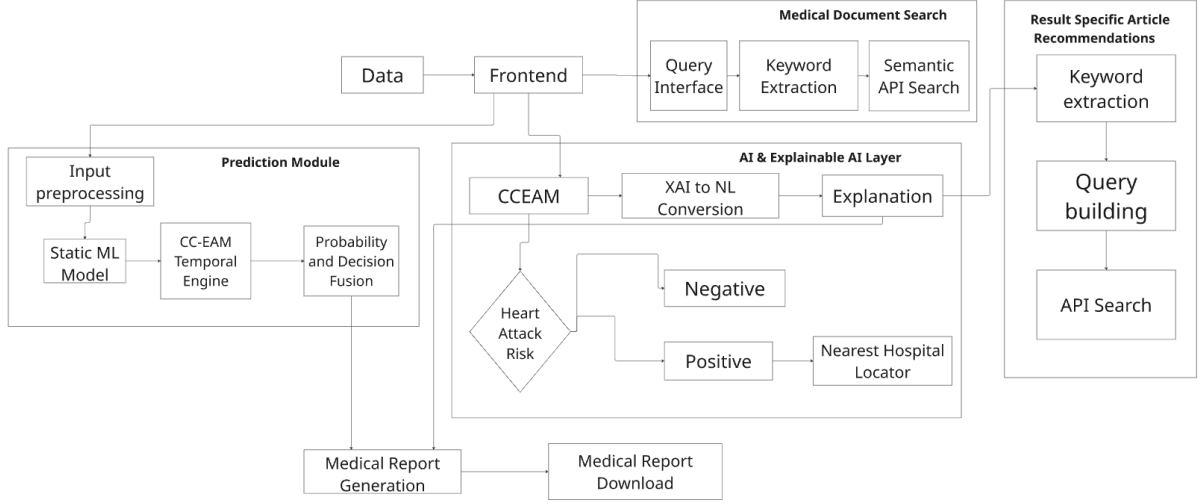


Figure 3.1: Overall System Architecture

The architecture begins with patient data acquisition and preprocessing, followed by the risk prediction engine that integrates static machine learning models and the CC-EAM model. The Explainable AI layer interprets model predictions and generates human-readable explanations. Additional modules handle report generation, hospital recommendations, and medical literature retrieval.

3.4 Module-wise Architecture and Description

The system is divided into several functional modules that work together to perform risk prediction and generate actionable insights.

3.4.1 Input Preprocessing Module

This module prepares the raw input data for machine learning analysis.

- Accepts patient biomarker inputs such as Age, Troponin, and CK-MB.
- Performs data cleaning to handle missing or inconsistent values.
- Applies normalization and scaling techniques to maintain consistency with the training data.
- Converts categorical variables into encoded representations where required.

This preprocessing step ensures that the input data is compatible with the trained machine learning models.

3.4.2 Risk Prediction Module

The risk prediction module is responsible for estimating the probability of myocardial infarction.

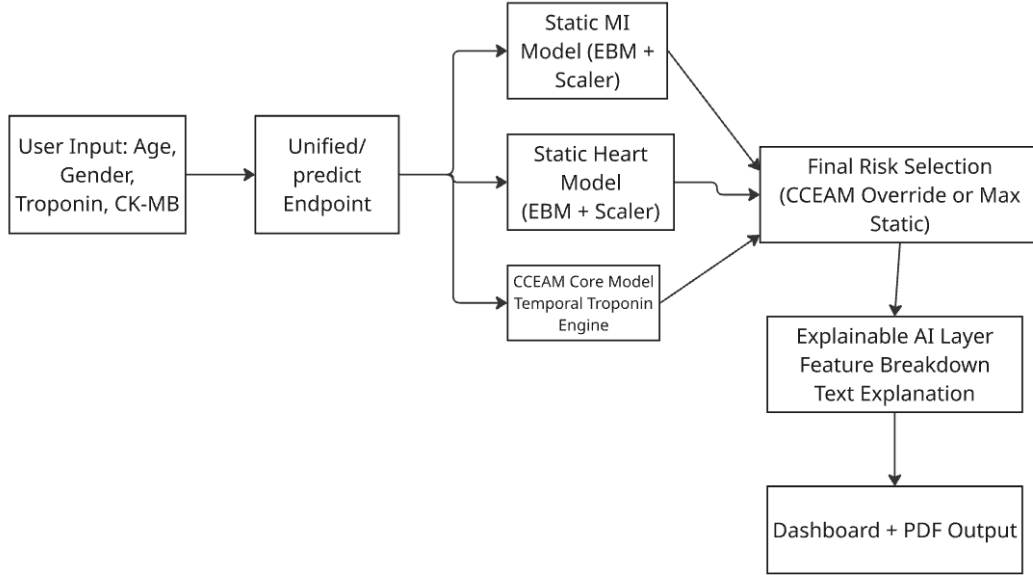


Figure 3.2: Risk Prediction Module

The module integrates multiple predictive components:

- Static myocardial infarction prediction model
- Static cardiovascular risk model
- CC-EAM temporal biomarker model

The models generate probability scores which are combined using a probability fusion mechanism. If acute biomarker indicators are present, the CC-EAM prediction overrides the static models to provide a more accurate diagnosis.

3.4.3 Clinically Constrained Explainable Additive Model (CC-EAM)

The CC-EAM model forms the core prediction engine of the system.

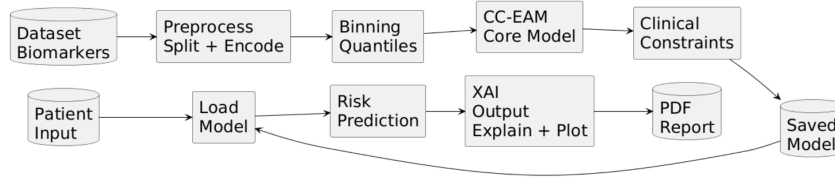


Figure 3.3: Clinically Constrained Explainable Additive Model

The model operates using the following process:

- Historical biomarker datasets are used for training.
- Continuous biomarker values are divided into quantile bins to capture nonlinear relationships.
- Each biomarker contributes independently to the final prediction score.
- Clinical constraints are applied to ensure medically valid relationships between biomarkers and risk levels.

The final prediction is obtained by summing the individual feature contributions and applying a logistic transformation to produce a probability score.

3.4.4 Explainable AI Module

The Explainable AI module provides transparency for the model predictions.

- Identifies feature contributions influencing the prediction.
- Generates visual explanations and contribution plots.
- Converts technical explanations into human-readable natural language descriptions.

This module ensures that clinicians can clearly understand why a specific prediction was made.

3.4.5 Medical Report Generation Module

The report generation module automatically creates a structured diagnostic report.

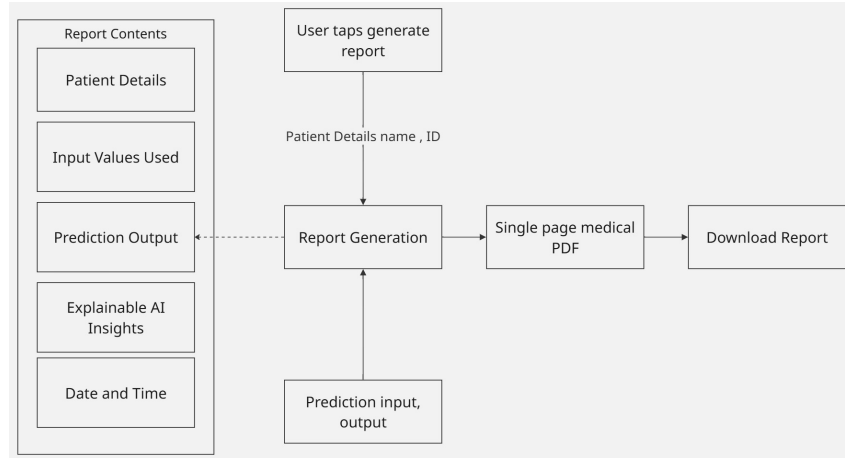


Figure 3.4: Report Generation Module

The generated report includes:

- Patient details and biomarker values
- Predicted myocardial infarction probability
- Risk category classification
- Explainable AI insights
- Date and time of analysis

The report is exported as a downloadable PDF document for clinical reference.

3.4.6 Medical Literature Recommendation Module

The Medical Literature Recommendation Module retrieves relevant research articles to support clinical interpretation of the predicted cardiac risk. This module enables clinicians to access evidence-based medical knowledge related to the patient’s biomarker profile and predicted risk category.

The module is implemented using the PubMed database through the NCBI Entrez API. It automatically constructs search queries based on biomarker abnormalities and predicted risk levels, allowing the system to retrieve research articles relevant to the clinical scenario.

- The system extracts structured metadata from PubMed articles including the title, authors, journal name, publication year, and abstract.

- Sentence-BERT embeddings are used to represent both the user query and research articles in a semantic vector space.
- Cosine similarity is applied to measure the semantic relevance between the generated query and retrieved research papers.
- A hybrid ranking strategy is implemented in which recent publications are prioritized first, followed by semantic relevance scores.

By combining semantic search with recency-based prioritization, the module ensures that clinicians receive both relevant and up-to-date medical literature corresponding to the predicted cardiac risk profile.

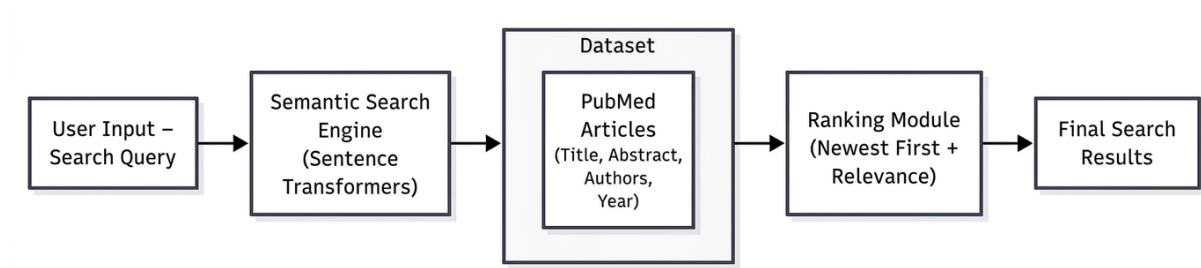


Figure 3.5: Architecture of the Medical Literature Recommendation Module

3.5 Summary

The proposed system integrates machine learning, explainable AI, and medical knowledge retrieval to create a comprehensive decision-support platform for heart attack prediction. By combining biomarker-based risk analysis with clinically constrained modeling, the system provides accurate predictions along with transparent explanations. The modular architecture ensures scalability and allows each component to operate independently while contributing to the overall diagnostic workflow.

Chapter 4

System Design

This chapter presents the overall design and architecture of the proposed AI-based myocardial infarction detection system. It explains how various components, including machine learning models, Explainable AI methods, and the user interface, integrate to form a complete and functional framework. The chapter also outlines the algorithms used, software and hardware requirements, dataset details, module division, and project workflow, providing a clear view of how the system is structured and implemented.

4.1 Overall System Architecture

The proposed system architecture is designed in multiple layers to ensure seamless data flow, efficient processing, and clear interpretability. Each layer has a distinct role, collectively forming a robust and explainable decision support system for myocardial infarction detection. The major layers include the **Database Layer**, **Backend (Model & Logic Layer)**, **Explainable AI Layer**, and **Frontend (User Interface)**.

4.1.1 Database Layer

Patient Records: This module stores all relevant patient data, which serves as the foundational data source for the entire system.

4.1.2 Backend (Model & Logic Layer)

This layer handles data processing, model development, prediction, and explainability.

- **Data Preprocessing & Validation:** Cleans, transforms, normalizes, and prepares the raw data from the Patient Records for model training. It also validates the data for quality and consistency.

- **Feature Engineering:** Selects, creates, or modifies relevant features from the preprocessed data to improve the performance of the machine learning models.
- **Model Training & Selection (LR, RF, XGBoost, EBM, ANN):** Trains various machine learning models such as Logistic Regression (LR), Random Forest (RF), XGBoost, Explainable Boosting Machines (EBM), and Artificial Neural Networks (ANN). The best performing model is selected based on evaluation metrics.
- **Prediction Engine:** The core module that receives patient data (either from the database or via the Patient Data Entry Form) and uses the final, trained model (or ensemble) to output the heart attack detection result.

4.1.3 Explainable AI Layer

Provides interpretability for the predictions made by the Prediction Engine. It uses Explainable AI (XAI) techniques like SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and insights derived from EBM to explain why a particular prediction was made. This is crucial for building trust with clinicians.

4.1.4 Frontend (User Interface)

This layer represents the main interaction point for clinicians and includes visualization tools and decision-support interfaces.

- **Clinician:** The primary user of the system, typically a doctor or healthcare professional, who interacts with the frontend.
- **Patient Data Entry Form:** An interface that allows the clinician to manually input new or updated patient data (e.g., vital signs, lab results) for immediate analysis by the Prediction Engine.
- **Prediction Output Dashboard:** Displays the results from the Prediction Engine, showing the heart attack risk or early warning status in a clear, summarized format. It acts as the central hub for viewing the system's core output.

- **Explanation (XAI) View:** Visualizes the explanations generated by the Explainable AI Layer. It shows the factors or features (e.g., blood pressure, age) that most influenced the prediction, helping the clinician understand the model's reasoning.
- **Mini Medical Knowledge Search:** A utility within the frontend that allows the clinician to search for medical information directly, potentially related to the prediction or patient condition.
- **External Knowledge APIs (eg:PubMed):** Provides the data source for the Mini Medical Knowledge Search, integrating access to external, peer-reviewed medical literature and research papers (such as those from PubMed) to support clinical decision-making.

4.2 Design Diagrams

4.2.1 Use Case Diagram

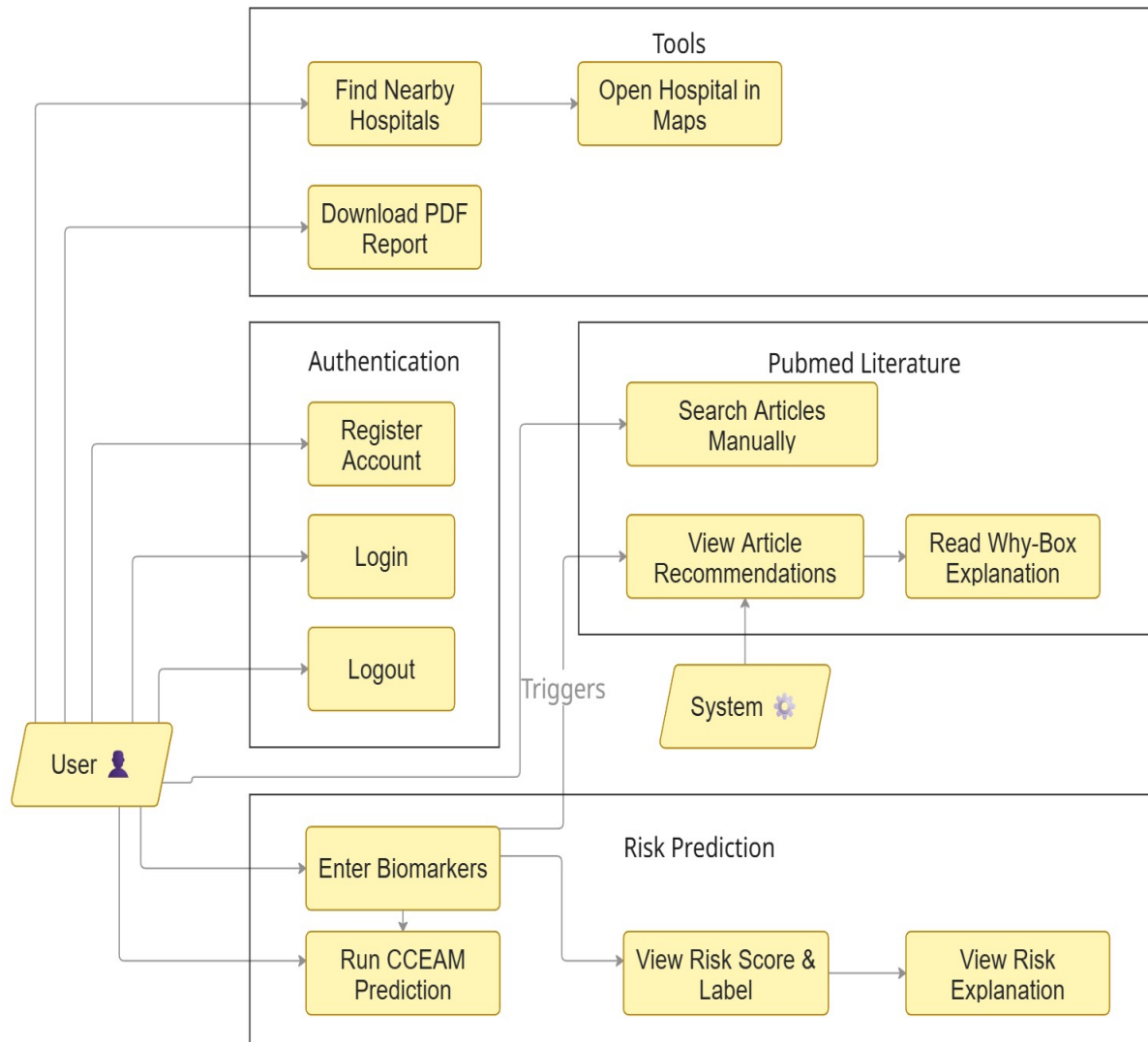


Figure 4.1: Use Case Diagram

The use case diagram illustrates interactions between key users—Clinician, Admin, and Common User—and the AI-Based Myocardial Infarction Detection System. Clinicians input patient biomarker data, request MI predictions, and view explainable results using SHAP and LIME. Admins manage training datasets and model updates to maintain system accuracy, while common users can access medical knowledge resources. Internal processes such as data preprocessing, model execution, and explanation generation work

together within the Machine Learning and XAI layer to ensure accurate, interpretable, and reliable diagnostic support.

4.2.2 Sequence Diagram

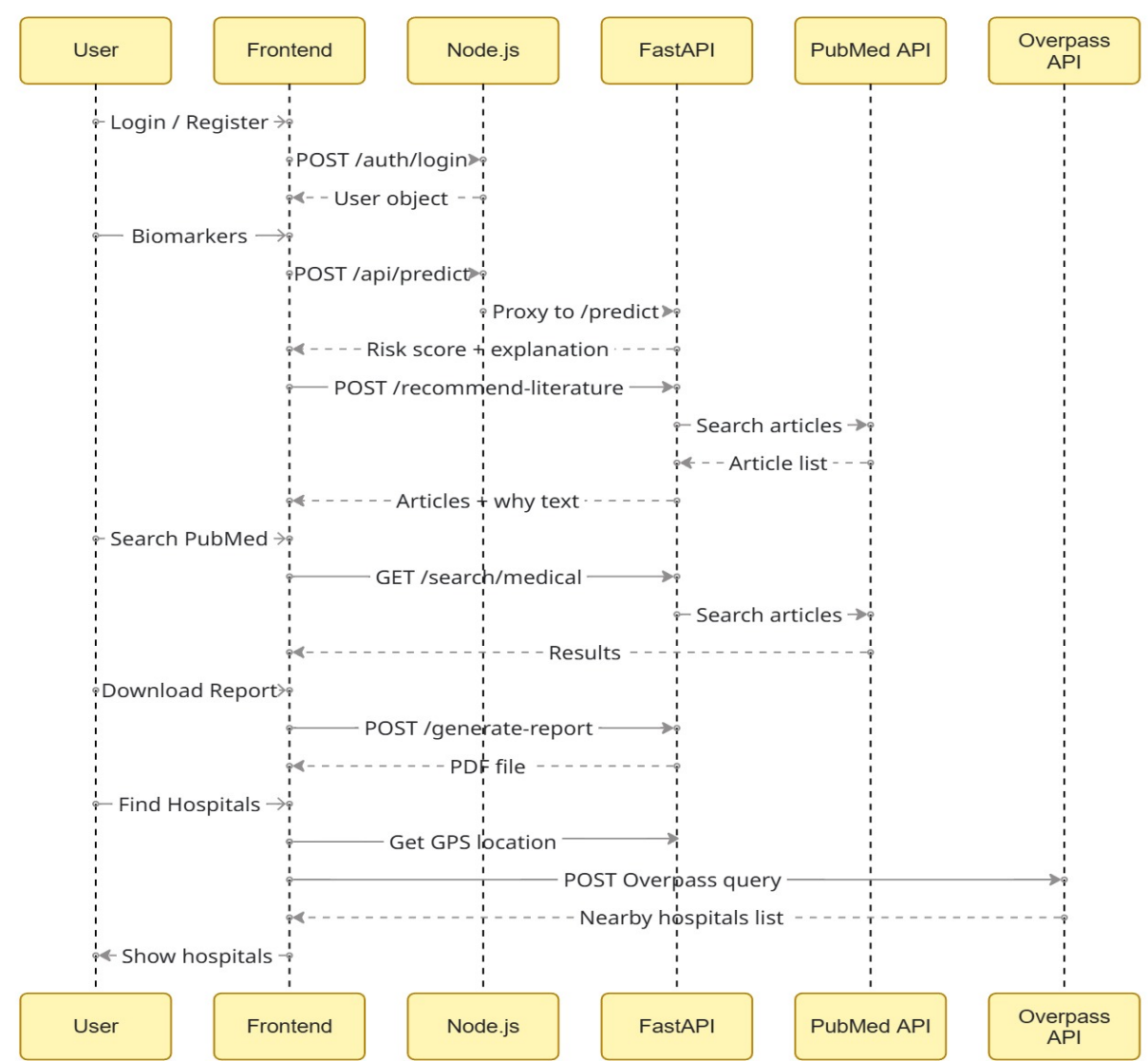


Figure 4.2: Sequence Diagram

The sequence diagram represents the workflow of the AI-based myocardial infarction detection system involving two primary users — Clinician and Common User (General Public).

Both users interact with the Frontend UI, which collects biomarker or health data such as Troponin, CK-MB, and CRP. The frontend sends this data to the Backend (Model

Logic Layer), where data preprocessing, normalization, and validation occur. The back-end retrieves reference or training data from the Database/Knowledge Base and sends preprocessed data to the Prediction + XAI Engine.

Within this engine, trained ML models (Logistic Regression, Random Forest, XG-Boost, and EBM) generate MI predictions and confidence scores. Explainable AI methods are applied to provide feature importance and interpretability. The results — including prediction and explanation — are sent back to the frontend.

Clinicians view a detailed diagnostic dashboard with explanations, while common users receive a simplified view of prediction outcomes. Clinicians can also search medical databases for relevant research and guidelines. The system ensures data security (HIPAA/GDPR compliance) and supports ensemble voting and risk stratification for improved decision accuracy.

4.3 Algorithms Used

This section describes the machine learning algorithms employed in the proposed system for myocardial infarction detection. Each algorithm contributes differently in terms of accuracy, robustness, and interpretability. The combination of these models enables both high performance and clinical transparency.

4.3.1 Algorithm 1: Logistic Regression (LR)

Algorithm 1 Logistic Regression (LR) Algorithm

Input: Training data (X, y) , learning rate η , iterations N

- 1: Initialize weights $w = [w_0, w_1, \dots, w_n]$ randomly
- 2: **for** $i = 1$ to N **do**
- 3: Compute linear combination: $z = w_0 + \sum_{j=1}^n w_j x_j$
- 4: Apply sigmoid function: $\hat{y} = \frac{1}{1+e^{-z}}$
- 5: Compute loss: $L = -\frac{1}{m} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$
- 6: Update weights: $w_j = w_j - \eta \frac{\partial L}{\partial w_j}$
- 7: **end for**

Output: Optimized weights w , class probabilities $\hat{y}$

4.3.2 Algorithm 2: Support Vector Machine (SVM)

Algorithm 2 Support Vector Machine (SVM) Algorithm

Input: Training data (x_i, y_i) , where $y_i \in \{-1, +1\}$

- 1: Initialize weight vector w and bias b
- 2: Define decision function $f(x) = w \cdot x + b$
- 3: Solve optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad s.t. \quad y_i(w \cdot x_i + b) \geq 1$$

- 4: Use Lagrange multipliers to obtain optimal w and b
- 5: For nonlinear data, apply kernel trick (e.g., RBF, polynomial)
- 6: Predict class: $\hat{y} = \text{sign}(w \cdot x + b)$

Output: Decision boundary parameters w, b

4.3.3 Algorithm 3: Random Forest (RF)

Algorithm 3 Random Forest (RF) Algorithm

Input: Dataset D , number of trees T

- 1: **for** $t = 1$ to T **do**
- 2: Draw a bootstrap sample D_t from D
- 3: Select a random subset of features F_t
- 4: Build a decision tree $h_t(x)$ using D_t and F_t
- 5: **end for**
- 6: For a new instance x , obtain predictions from all trees:
- 7: $\hat{y}_t = h_t(x), \quad t = 1, 2, \dots, T$
- 8: Aggregate predictions:
- 9: $\hat{y} = \text{majority vote (classification) or mean}(\hat{y}_t) \text{ (regression)}$

Output: Final ensemble prediction $\hat{y}$

4.3.4 Algorithm 4: XGBoost (Extreme Gradient Boosting)

Algorithm 4 Extreme Gradient Boosting (XGBoost) Algorithm

Input: Training data (X, y) , number of trees T , learning rate η

- 1: Initialize model prediction: $\hat{y}_i^{(0)} = \text{mean}(y)$
- 2: **for** $t = 1$ to T **do**
- 3: Compute gradients: $g_i = \frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}$
- 4: Fit regression tree $f_t(x)$ to predict g_i
- 5: Compute optimal leaf weights using second derivative (Hessian)
- 6: Update predictions: $\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i)$
- 7: **end for**

Output: Final prediction $\hat{y} = \sum_{t=1}^T \eta f_t(x) = 0$

4.3.5 Algorithm 5: Explainable Boosting Machine (EBM)

Algorithm 5 Explainable Boosting Machine (EBM) Algorithm

Input: Dataset (X, y) , learning rate η , iterations N

- 1: Initialize model $f(x) = \text{bias}$
- 2: **for** $i = 1$ to N **do**
- 3: **for** each feature x_j **do**
- 4: Fit a small tree $f_j(x_j)$ to predict residuals
- 5: Update model: $f(x) = f(x) + \eta f_j(x_j)$
- 6: **end for**
- 7: Optionally add pairwise interactions $f_{ij}(x_i, x_j)$
- 8: **end for**
- 9: Final model:

$$f(x) = \text{bias} + \sum_j f_j(x_j) + \sum_{i < j} f_{ij}(x_i, x_j)$$

- 10: Predict for classification: $\hat{y} = \sigma(f(x))$

Output: Explainable model with additive feature contributions $= 0$

4.3.6 Algorithm 6: CCEAM (Clinically Constrained Explainable Additive Model)

Algorithm 6 CCEAM Algorithm

Input: Dataset D , features $F = \{Age, Gender, Troponin, CK_MB\}$, number of bins K , learning rate η , number of epochs E

Initialize

- 1: Initialize bias term β_0
- 2: Initialize feature weights w_f for all $f \in F$
- 3: Compute quantile bin edges for continuous features

Training Phase

- 4: **for** $epoch = 1$ to E **do**
- 5: **for** each training instance (x_i, y_i) in dataset D **do**
- 6: Determine bin indices for all features
- 7: Compute additive logit
- 8: $s_i = \beta_0 + \sum_{f \in F} w_f[k_f]$
- 9: Compute probability $\hat{p}_i = \sigma(s_i)$
- 10: Update model parameters using gradient descent
- 11: **end for**
- 12: Enforce monotonic increasing constraints on Age, Troponin, and CK-MB
- 13: **end for**

Prediction and Explanation

- 14: For a query input x , compute feature contributions c_f
- 15: Compute MI probability
- 16: $\hat{p} = \sigma(\beta_0 + \sum_f c_f)$
- 17: Generate explanation
- 18: $\mathcal{E} = \{(f, x_f, c_f)\}_{f \in F}$

Output: Predicted MI probability $\hat{p}$ and explanation $\mathcal{E}$

4.4 Software and Hardware Requirements

This section lists the essential software tools, libraries, and hardware configurations required for developing and running the proposed AI-based myocardial infarction detection system.

4.4.1 Software Requirements

- **Programming Language:** Python 3.x
- **Libraries and Frameworks:**
 - **Data Handling and Analysis:** pandas, numpy
 - **Visualization:** matplotlib, seaborn
 - **Model Development:** scikit-learn
 - **Deep Learning (optional):** TensorFlow or PyTorch
- **Development Environment:** Jupyter Notebook
- **Frontend Interface:** Streamlit for interactive user interface and deployment

4.4.2 Hardware Requirements

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or higher
- **Memory (RAM):** Minimum 8 GB
- **CPU Cores:** Minimum 4-core processor
- **Operating System:** Windows 10/11, Ubuntu 20.04+, or macOS

4.5 Dataset Identified

Two primary datasets were utilized in the development of the AI-based myocardial infarction detection system. Each dataset contains patient-specific biomarkers and demographic details that serve as inputs for model training and evaluation. These datasets help the system learn patterns related to heart attack risk and improve predictive accuracy.

4.5.1 Dataset

- **Data Fields:** Age, Gender, Troponin, CK-MB
- **Number of Entries:** 10,545
- **Description:** This dataset focuses on key cardiac biomarkers such as Troponin and CK-MB, which are critical indicators of myocardial infarction. It is primarily used for baseline model training and validation.

4.6 Project Timeline

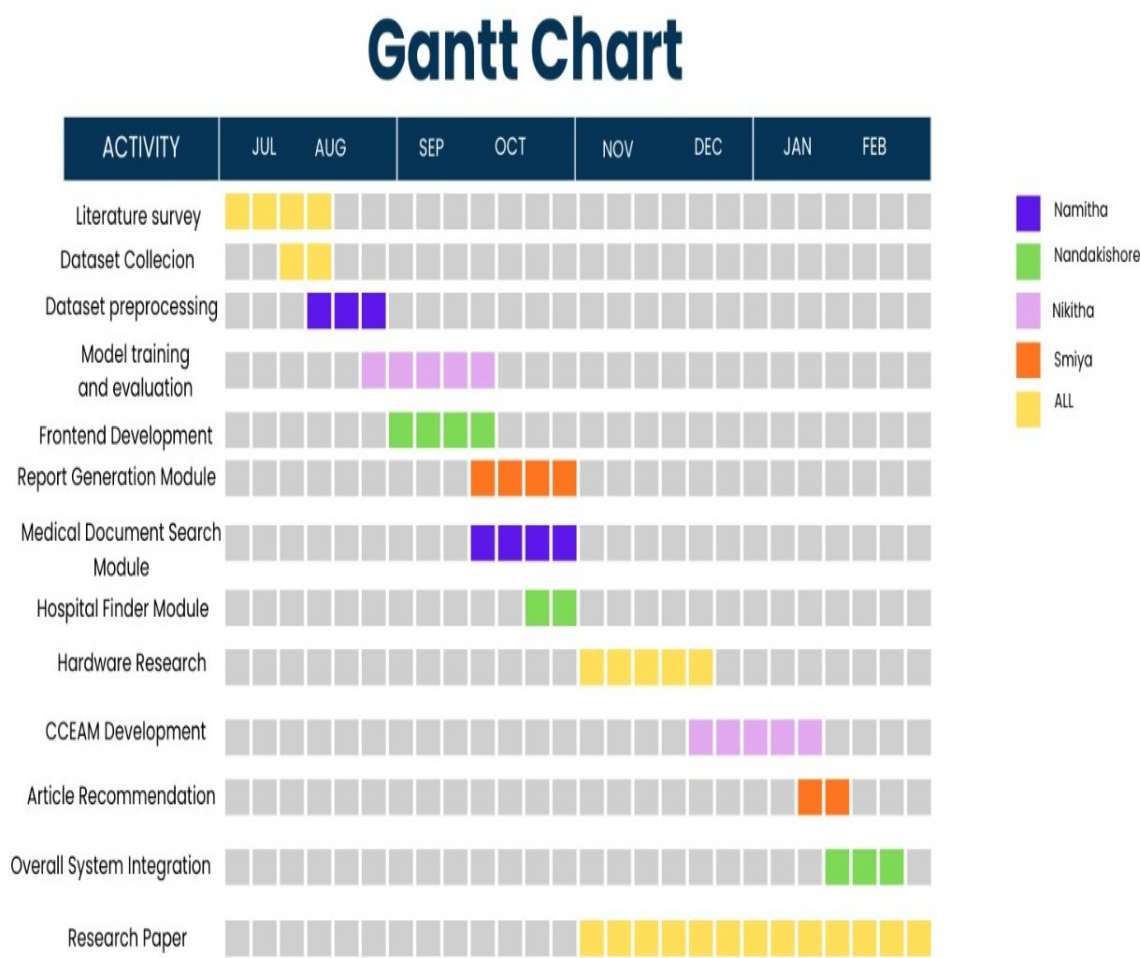


Figure 4.3: Project Timeline (Gantt Chart)

4.7 Module Division and Work Responsibilities

The project tasks are divided among the team members based on their technical expertise and focus areas. This distribution ensures efficient collaboration, modular development, and smooth integration of components. The work division is summarized in Table 4.1.

Table 4.1: Work Division Chart

Sl. No.	Module / Task	Team Member	Description
1	Literature Survey & Research	All Members	Conducted background study on MI prediction, biomarker-based diagnosis, explainable AI techniques, and existing clinical decision support systems.
2	Dataset Collection & Preprocessing	Namitha Mariam John	Responsible for collecting biomarker datasets, cleaning missing values, handling class imbalance using SMOTE, performing feature selection, and preparing structured input data for model training.
3	Machine Learning Model Development & Evaluation	Nikitha Sarah John	Implemented machine learning algorithms such as Random Forest, XG-Boost, Logistic Regression, and SVM. Performed model training, hyperparameter tuning, cross-validation, and performance comparison.
4	Medical Document Search Module	Namitha Mariam John	Designed and implemented the intelligent medical literature search module for retrieving relevant research papers and clinical references.

Sl. No.	Module / Task	Team Member	Description
5	Frontend Development	Nandakishore Suresh Nair	Developed the user interface for patient data input, prediction visualization, and interactive dashboard display.
6	Hospital Finder Module	Nandakishore Suresh Nair	Implemented hospital search functionality to identify nearby cardiac care centers for emergency medical assistance.
7	CCEAM Development	Nikitha Sarah John	Developed an additive model for heart attack risk detection to enhance prediction interpretability and accuracy.
8	Report Generation Module	Smiya Shaji	Developed an automated medical report generation system based on prediction results and biomarker analysis.
9	System Integration	Nandakishore Suresh Nair	Integrated backend machine learning models, explainable AI components, user interface, and additional modules into a unified working system.
10	Research Paper & Documentation	All Members	Prepared technical documentation, maintained reports, and drafted the research paper for publication.

4.8 UI Design

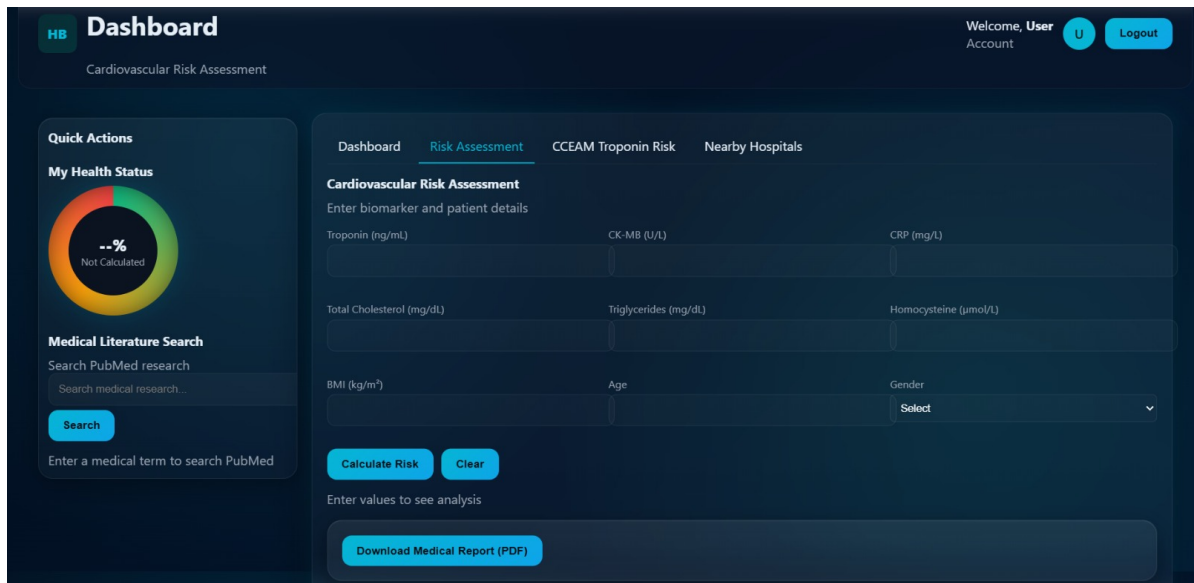


Figure 4.4: UI Design - Overview Screen

4.9 Expected Outcomes

The proposed AI-based myocardial infarction detection system is expected to deliver both technical and clinical benefits. By integrating multiple machine learning algorithms with explainable AI techniques, the system aims to provide accurate, interpretable, and clinically relevant results that can assist healthcare professionals in decision-making.

- **Accurate Prediction:** The system will predict whether a patient is likely affected by myocardial infarction based on biomarker data, offering high accuracy and reliability.
- **Explainable Results:** Through local explanations and visualizations, the system will provide clear explanations for each prediction, enabling clinicians to understand the reasoning behind model outputs.
- **Comparative Model Analysis:** Evaluation and comparison of multiple algorithms (LR, RF, XGBoost, SVM, EBM, CCEAM) will highlight the most effective model in terms of performance and interpretability.

- **Interactive Clinical Dashboard:** A user-friendly frontend built with Streamlit will allow clinicians to input patient data, view predictions, and explore explanation visualizations in real time.
- **Knowledge Integration:** The inclusion of a mini medical knowledge search tool connected to external APIs (e.g., PubMed) will support evidence-based medical decision-making.
- **Research Contribution:** The project will contribute to medical AI research by demonstrating a transparent and reproducible framework for integrating explainable AI in clinical diagnosis.

Overall, the system is expected to enhance diagnostic accuracy, improve clinician confidence through transparency, and pave the way for intelligent, trustworthy healthcare tools.

4.10 Summary of Chapter

This chapter presented the detailed system design of the proposed AI-based myocardial infarction detection system. It outlined the overall architecture, key components, algorithms used, and the software–hardware setup required for implementation. The chapter also described how various modules—including data preprocessing, model training, prediction engine, and explainable AI layer—interact to form an integrated and interpretable diagnostic framework.

The datasets identified were discussed along with their structure and relevance to the problem domain. Furthermore, the expected outcomes of the system were highlighted, emphasizing improved diagnostic accuracy, interpretability, and clinical usability.

Overall, this chapter provided a comprehensive view of how the conceptual framework is translated into a practical design, forming the foundation for the subsequent implementation and evaluation phases of the project.

Chapter 5

Results and Discussions

5.1 Overview

This chapter presents the results obtained from the implementation of the proposed AI-driven heart attack prediction and detection system. The outputs generated by the CC-EAM model are analyzed to evaluate the system's ability to identify potential cardiac risks using biomarker inputs such as troponin and CK-MB along with demographic information.

The chapter demonstrates how the developed system processes user-provided biomarker data, performs prediction using trained machine learning models, and produces interpretable outputs using Explainable AI techniques. The results include prediction outputs, explanation visualizations, and the functionality of additional modules integrated into the application.

Furthermore, this chapter discusses the effectiveness of the system in terms of prediction capability, interpretability, and usability. The outputs presented in the following sections illustrate how the model supports decision-making by providing clear insights into cardiac risk levels and assisting users with relevant medical information and emergency support features.

5.2 Results

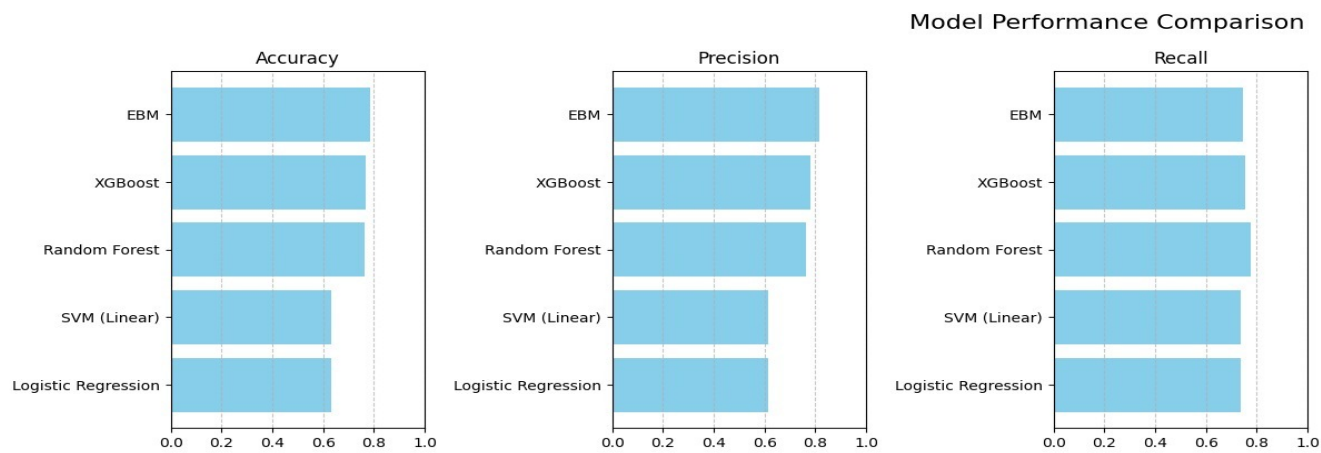


Figure 5.1: Model Performance Comparison 1

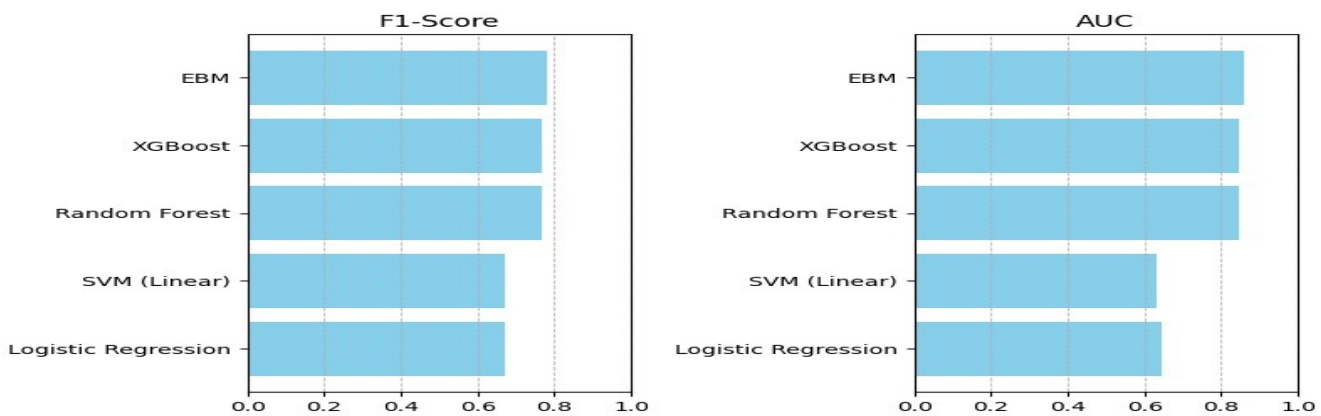


Figure 5.2: Model Performance Comparison 2

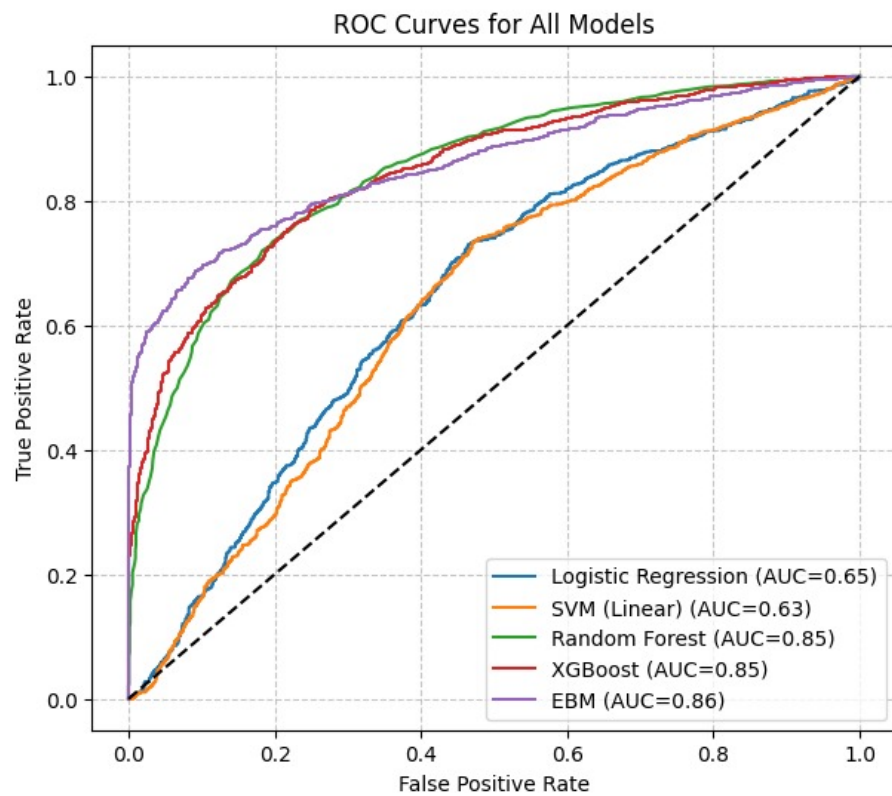


Figure 5.3: ROC Curves for all Models

```

--- CC-EAM Model Evaluation ---
Accuracy: 0.9597
Precision: 0.9623
Recall: 0.9559
F1-score: 0.9591
ROC-AUC: 0.9911

```

Figure 5.4: CCEAM model evaluation

5.3 Output

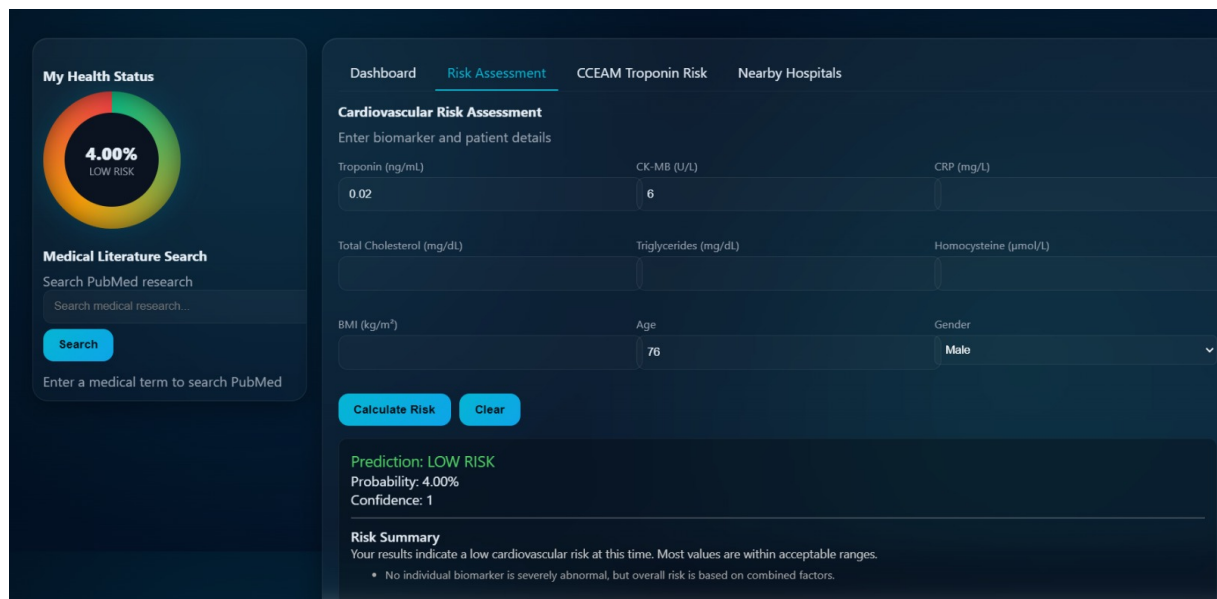


Figure 5.5: Risk Assessment Dashboard

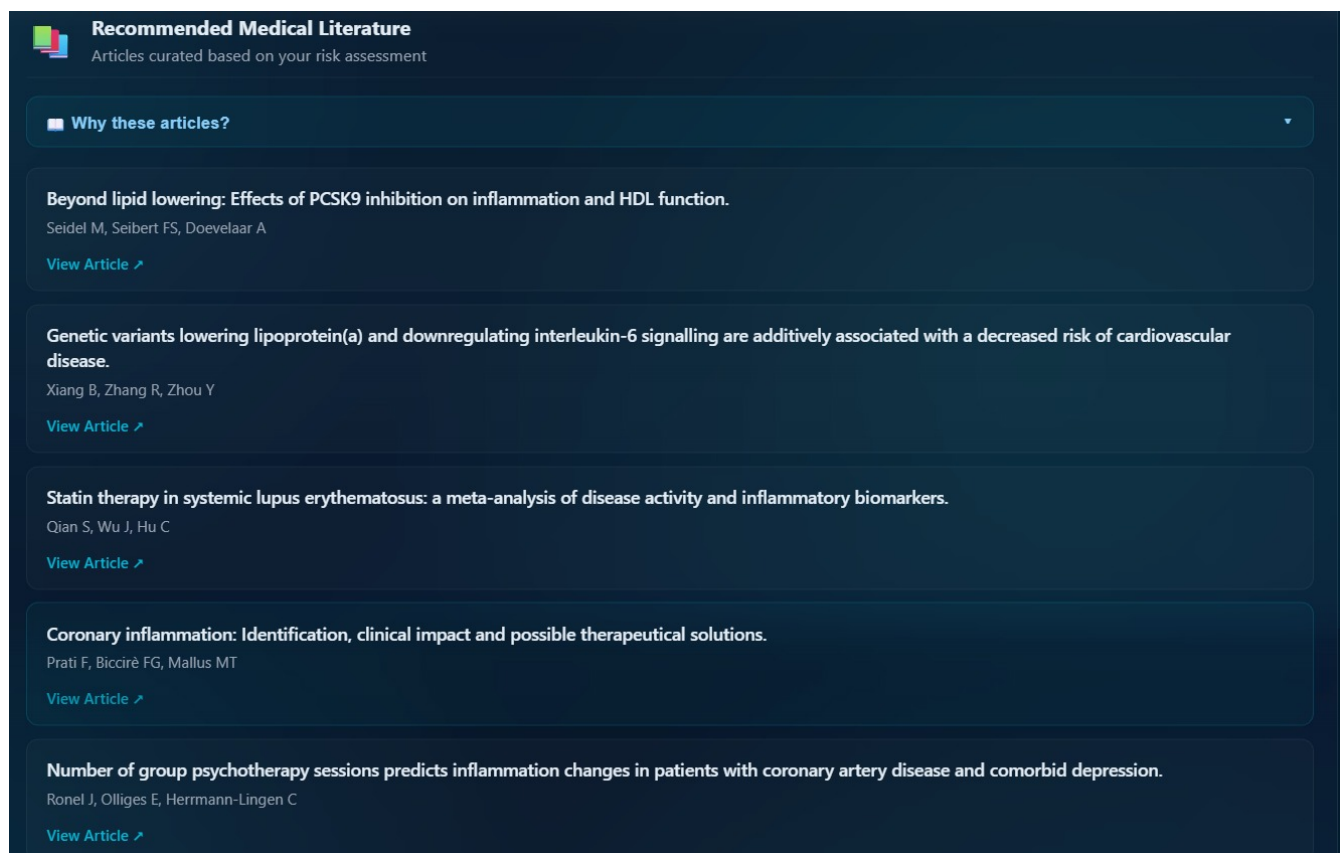


Figure 5.6: Recommended Medical Literature

DashboardRisk AssessmentCCEAM Troponin RiskNearby Hospitals

CCEAM Acute MI Risk

Temporal troponin prediction

Age

49

Gender

Male

Troponin 0h (ng/mL)

0.1

CK-MB (U/L)

23

Predict Risk

Download CCEAM Report (PDF)

Prediction: HIGH RISK

Probability: 99.00%

Troponin 1h: 0.14

Troponin 2h: 0.18

Figure 5.7: CCEAM Troponin Risk (High)

DashboardRisk AssessmentCCEAM Troponin RiskNearby Hospitals

CCEAM Acute MI Risk

Temporal troponin prediction

Age

23

Gender

Male

Troponin 0h (ng/mL)

0.002

CK-MB (U/L)

23

Predict Risk

Download CCEAM Report (PDF)

Prediction: LOW RISK

Probability: 1.44%

Troponin 1h: 0.00

Troponin 2h: 0.00

Blood Biomarker ML Dashboard • Secure Health Analytics

Figure 5.8: CC-EAM Troponin Risk (Low)

Myocardial Infarction (MI) Prediction Report

Patient Details

Age: 36.0
Gender: Male
Troponin: 0.08
CK-MB: 2.0

Prediction Output

Predicted MI Risk: 99.00% (HIGH)

Medical Interpretation

High probability of acute myocardial infarction (MI). Biomarkers strongly suggest myocardial injury. Immediate clinical evaluation is recommended.

Summary

The model predicts a high chance of heart attack. The results look serious and urgent medical attention is needed.

Explainable AI Breakdown (Feature Contributions)

Feature	Value	Contribution
Age	36.0	0.3015
Gender	1	0.8841
CK_MB	2.0	1.5049
Troponin	0.08	2.6308
Troponin_1h	0.10907	3.5867
Troponin_2h	0.14534	4.7795

Top Reasons

Predicted MI Risk: 99.00% (HIGH risk) Main factors increasing risk: - Troponin_2h (value=0.14534) increased risk - Troponin_1h (value=0.10907) increased risk - Troponin (value=0.08) increased risk

Reference Ranges

Troponin (0h, 1h, 2h): Normal < 0.04 ng/mL
CK-MB: Normal < 5 U/L

Figure 5.9: MI Prediction Report

Nearby Heart Hospitals

Find cardiology hospitals near your location.

[Find Hospitals Near Me](#)

Ambalamugal Medical Aid Society (AMAS)

 Address not available
 Not available
3.31 km away
[Open in Map](#)

Urban Primary Health Centre, Chalikkavattom

 Address not available
 Not available
3.96 km away
[Open in Map](#)

Sunrise Hospital, Ernakulam

 Address not available
 Not available
4.15 km away
[Open in Map](#)

Nirmala Medical Mission Hospital

 Address not available
 Not available
4.43 km away

Figure 5.10: Nearby Hospital Recommendations

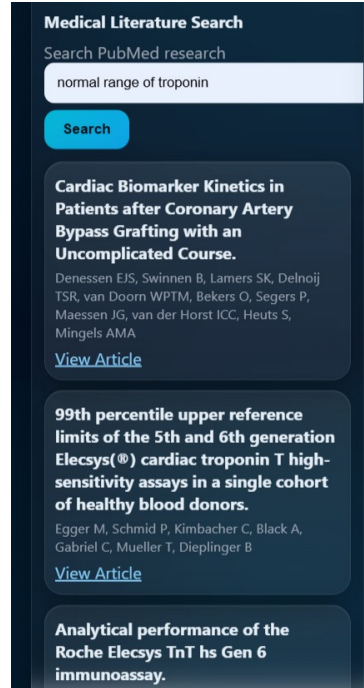


Figure 5.11: Medical Literature Search

5.4 Discussion

The results obtained from the system demonstrate that the CC-EAM model can effectively analyze biomarker data to identify possible signs of cardiac risk. By processing important indicators such as troponin and CK-MB, the system is capable of detecting abnormal biomarker patterns that may indicate the presence of acute myocardial infarction or increased heart attack risk.

A significant advantage of the proposed system is the integration of Explainable AI, which enhances the interpretability of predictions. Instead of providing only a risk score, the system explains how individual biomarkers contribute to the final prediction. This transparency improves user trust and makes the results easier to understand for both users and healthcare professionals.

The developed application also demonstrates the successful integration of supportive features such as the medical research search engine, which retrieves information from verified scientific journals, and the hospital locator service, which helps users identify nearby medical facilities during emergencies. These components extend the functionality of the system beyond prediction and make it a comprehensive decision-support tool.

Overall, the results indicate that the proposed system provides meaningful insights into

cardiac risk assessment. However, further improvements such as the use of larger clinical datasets, real-world validation, and integration with healthcare infrastructure could enhance the reliability and practical deployment of the system.

5.5 Summary

This chapter presented the results obtained from the implementation of the proposed AI-based heart attack prediction and detection system. The outputs demonstrate the capability of the CC-EAM model to analyze biomarker inputs and generate interpretable predictions related to cardiac risk.

The discussion highlights the importance of combining machine learning with Explainable AI to ensure transparency and reliability in medical prediction systems. Additionally, the integration of supporting modules such as the medical search engine and hospital locator service enhances the overall usability of the application.

Overall, the results confirm that the proposed system can function as an effective AI-assisted decision-support platform for early cardiac risk assessment. The chapter also emphasizes the need for further validation and development to improve the system's applicability in real-world healthcare settings.

Chapter 6

Conclusion

6.1 Conclusion

This project demonstrates the development of an AI-driven heart attack prediction and detection system that combines machine learning with Explainable AI (XAI) to support timely and informed medical decision-making. By analyzing critical cardiac biomarkers such as troponin and CK-MB, the proposed CC-EAM model is able to evaluate both immediate heart attack risk and potential cardiac abnormalities. Unlike traditional black-box models, the system provides transparent explanations for its predictions, allowing users and healthcare professionals to understand the reasoning behind the results.

The integration of additional features such as a medical research search engine, which retrieves information from verified scientific journals, and a location-based hospital finder, further enhances the practical usability of the system. These components transform the application from a simple prediction tool into a comprehensive decision-support platform for cardiac health awareness and response.

Overall, the proposed system highlights how artificial intelligence, when combined with interpretability and user-focused design, can contribute to early risk assessment and encourage timely medical intervention during potential cardiac events. With further validation using larger clinical datasets and integration with real-world healthcare systems, this approach has the potential to support preventive cardiology and improve patient outcomes.

6.2 Future Scope

- Development of a portable hardware device for real-time blood biomarker analysis.
- Expansion to multi-modal prediction using biomarkers + ECG + clinical history.
- Large-scale clinical validation across diverse populations.
- Integration with hospital systems as a Clinical Decision Support Tool.

6.3 Summary

This chapter summarizes the outcomes and implications of the proposed AI-driven heart attack prediction and detection system. It highlights how the developed CC-EAM model analyzes key cardiac biomarkers such as troponin and CK-MB to assess heart attack risk while incorporating Explainable AI to provide transparent and understandable predictions. The chapter also emphasizes the additional features of the system, including a medical research search engine and a location-based hospital finder, which enhance the application's usability as a comprehensive decision-support tool for cardiac health.

Furthermore, the chapter outlines potential future improvements, such as developing a portable biomarker detection device, integrating multiple health signals like ECG and clinical history, conducting large-scale clinical validation, and connecting the system with hospital infrastructures. These future directions aim to improve the reliability, scalability, and real-world applicability of the proposed solution in supporting early cardiac risk detection and preventive healthcare.

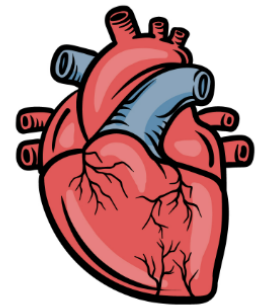
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 - [8] Emakhu, J., Monplaisir, L., Aguwa, C., *et al.* (2022) ‘Acute coronary syndrome prediction in emergency care: a machine learning approach’, *Computers in Biology and Medicine*, 145, p. 105462. Available at: <https://doi.org/10.1016/j.compbiomed.2022.105462>.
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Appendix A: Presentation

CC-EAM : A Predictive Model for Heart Attack Detection Using Blood Biomarkers



FINAL PROJECT PRESENTATION

Guide: Ms. Sherine Sebastian

DATE: 05/03/2026

Team Members:

Namitha Mariam John – U2203145

Nandakishore Suresh Nair – U2203146

Nikitha Sarah John – U2203162

Smiya Shaji – U2203195

Page 1

TABLE OF CONTENTS

- Problem statement
- Project objective
- Current Limitations
- Purpose and need
- Literature Survey
- Proposed method
- Architecture Diagram
- Sequence diagram
- Use case diagram
- Module Diagrams and descriptions
- Results and outputs
- Assumptions
- Work breakdown and responsibilities
- Hardware and software requirements
- Gantt Chart
- Risks and challenges
- Future Scope
- Conclusion
- References

Page 2



INTRODUCTION

- This project focuses on developing an AI-powered system for detection and prediction of heart attacks.
- The system uses blood biomarkers such as Troponin, CK-MB.
- Unlike traditional ECG-based diagnosis, which has limited sensitivity (~50% in early stages) biomarker analysis provides higher accuracy for early myocardial infarction detection.
- The system integrates Explainable AI (XAI) methods to ensure predictions are transparent and interpretable enhancing trust in clinical decision-making.

Page 3



PROBLEM STATEMENT

To develop an AI-driven heart attack detection and predictive system that utilizes key blood biomarkers like troponin and CKMB to accurately predict myocardial infarction risk, while incorporating Explainable AI (XAI) techniques to provide transparent and interpretable diagnostic insights.

Page 4



PROJECT OBJECTIVES

- To develop an AI-based system for early prediction of myocardial infarction (heart attack) using clinical biomarker data.
- To implement a Clinically Constrained Explainable Additive Model (CC-EAM) for accurate and clinically consistent risk prediction.
- To generate a probability-based MI Risk Score instead of only binary classification.
- To integrate Explainable AI (XAI) for interpreting how each biomarker contributes to the prediction.
- To design a simple interface that clearly displays the risk score, risk category, and explanation.

Page 5



LIMITATIONS OF EXISTING SYSTEMS

- Existing models like FRS, GRACE, TIMI, and QRISK are primarily statistical scoring systems.
- Most are designed for long-term (10-year) cardiovascular risk prediction.
- Limited focus on immediate myocardial infarction detection using biomarkers.
- Provide fixed categorical scores rather than personalized probability outputs.
- Limited adaptability to new datasets and modern AI optimization.
- Lack advanced explainability beyond traditional scoring tables.

Page 6



NEED AND PURPOSE OF PROPOSED SYSTEM

- Need for an AI-driven system focused specifically on early myocardial infarction prediction.
- Requirement for biomarker-based, clinically relevant risk estimation.
- Need to minimize false negatives in critical cardiac cases.
- Purpose to provide probability-based risk scoring instead of rigid categories.
- Integration of Explainable AI using the Clinically Constrained Explainable Additive Model (CC-EAM).
- To support clinicians with transparent and reliable decision assistance.

Page 7

Comparison of Literature Survey papers

❏ Paper Title	≡ Key Features Used	≡ Algorithms Applied	≡ Special Techniques	≡ Major Findings	≡ Contribution to Your...
1 Enhancing Heart Attack Prediction: Feature Identification from Multiparametric Cardiac Data Using Explainable AI	Age, demographics, ECG data, LDL/HDL, clinical parameters	Logistic Regression, KNN, SVM, Random Forest, ANN	SHAP, LIME (Explainable AI)	ANN achieved best performance among tested models	Provides benchmark ML models and validates biomarker-based prediction with XAI support
2 A Biomarker Discovery of Acute Myocardial Infarction Using Feature Selection and Machine Learning	Structured clinical variables and biomarkers	LR, Decision Trees, RF, GBM, SVM, Naïve Bayes, ANN, Deep Learning (CNN, RNN, DNN)	PCA, Genetic Algorithms, Recursive Feature Elimination (RFE)	Deep learning outperforms traditional ML for complex datasets	Guides advanced feature selection and model experimentation strategies
3 Machine Learning for Early Prediction of Acute Myocardial Infarction or Death in Acute Chest Pain Patients	Age, Sex, hs-cTnT, Glucose, Creatinine, Hemoglobin	Logistic Regression, ANN	Clinical calibration for very high sensitivity and NPV	Biomarkers alone can reliably predict AMI or death	Strong justification for troponin-based prediction in your system
4 Prognostic Utility of a Multi-Biomarker Panel in Patients with Suspected MI	29 biomarkers; NT-proBNP, Apo A-I, KIM-1 identified as strongest predictors	LASSO regression	Multi-biomarker panel combined with GRACE score	Selected biomarkers significantly improved risk stratification	Emphasizes importance of LASSO-based biomarker selection
5 Fair and Explainable MI Prediction: Feature Selection & Class Imbalance Correction	Clinically relevant biomarkers	ML models with fairness evaluation	Class imbalance correction, cost-sensitive learning, Explainable AI	Improved predictive performance, fairness, and interpretability	Supports XAI integration and imbalance handling in medical datasets

Page 8

PROPOSED METHOD

- Collect patient biomarker inputs (Troponin, CK-MB, Age and other clinical parameters) through the system interface.
- Perform input preprocessing including normalization, scaling, and missing value handling to ensure clean and reliable data.
- Feed the processed data into the trained Clinically Constrained Explainable Additive Model (CC-EAM).
- Apply the static ML model along with the CC-EAM temporal engine for structured clinical risk evaluation.
- Perform probability estimation and decision fusion to generate a robust final prediction.

Page 9

PROPOSED METHOD

- Compute a myocardial infarction (heart attack) risk probability score instead of only binary classification.
- Classify the patient as Low Risk (Negative) or High Risk (Positive) based on the risk threshold.
- If classified as High Risk, trigger an alert and provide nearest hospital recommendations for emergency care.
- Generate an explainable output using XAI techniques to identify key contributing biomarkers.
- Convert model explanations into human-readable natural language for doctors and patients.

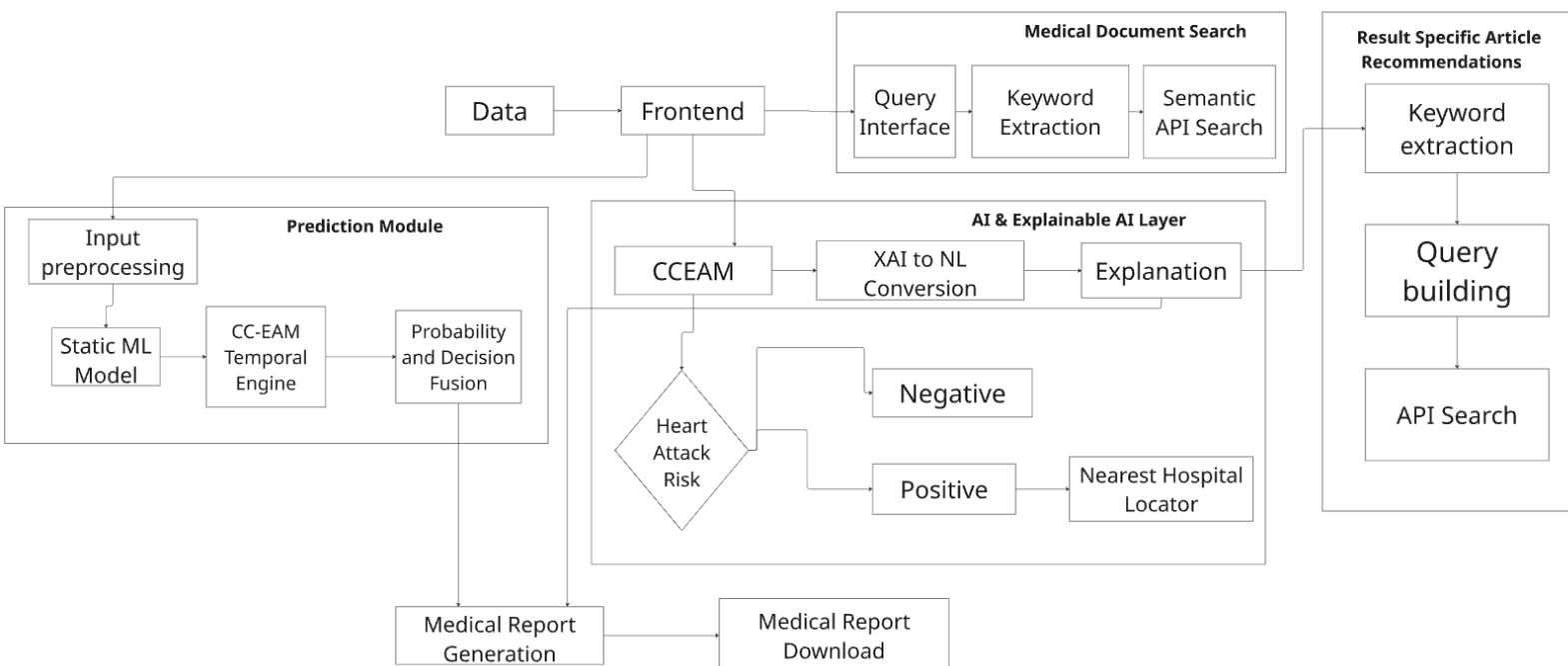
Page 10

PROPOSED METHOD

- Automatically generate a structured medical diagnostic report containing patient details, biomarker values, risk score, explanation, and recommended action.
- Enable secure download of the generated medical report.
- Extract important medical keywords from the prediction and explanation results.
- Construct intelligent medical search queries based on extracted keywords.
- Provide result-specific article recommendations to support clinical decision-making.

Page 11

ARCHITECTURE DIAGRAM

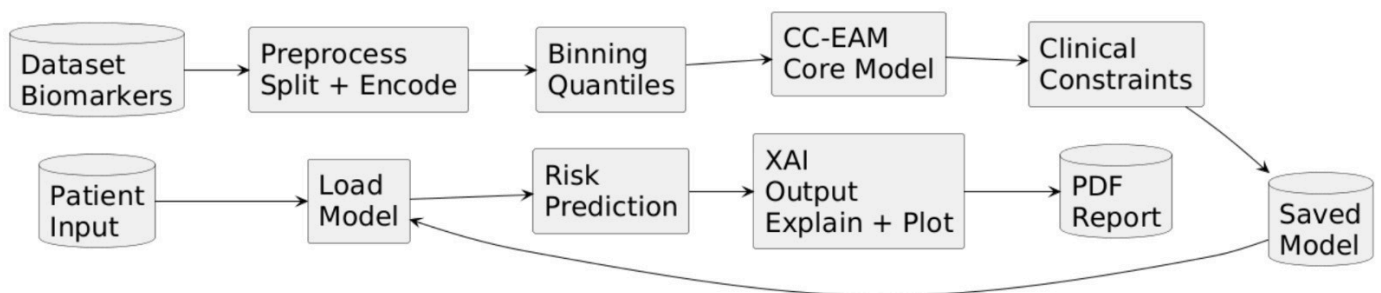


Page 12

MODULE - WISE DIAGRAMS AND EXPLANATIONS

Page 13

CC-EAM : Clinically Constrained Explainable Additive Model



Page 14

CC-EAM : Clinically Constrained Explainable Additive Model

- **Biomarker Dataset Input**

Historical patient biomarker data (Troponin, CK-MB, etc.) is collected.

- **Preprocessing (Split + Encode)**

Data is cleaned, encoded (if needed), and split into training & testing sets.

- **Quantile Binning**

Continuous biomarker values are divided into quantile bins to capture non-linear patterns and improve interpretability.

- **CC-EAM Core Training**

An Explainable Additive Model is trained where:

Each biomarker contributes independently

Final risk score = sum of individual feature contributions

- **Clinical Constraints Applied**

Medical rules (e.g., higher Troponin → higher risk) are enforced to ensure clinically valid predictions

Page 15

CC-EAM : Clinically Constrained Explainable Additive Model

- **Model Saved**

The trained and constrained model is stored for future use.

- **Patient Input (Deployment Phase)**

New patient biomarker values are entered into the system.

- **Risk Prediction**

Model predicts heart attack probability / risk category.

- **Explainable Output (XAI)**

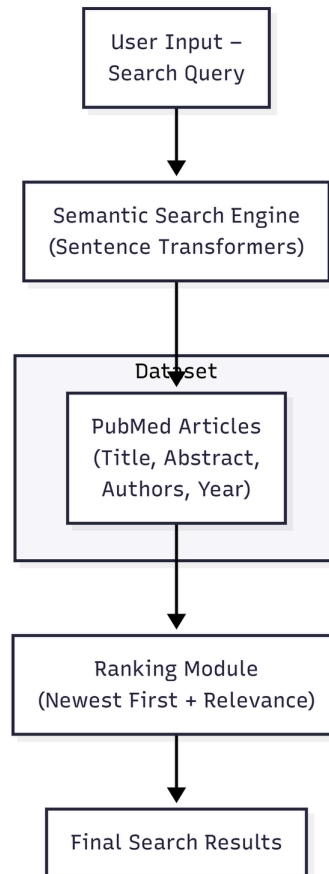
Feature contribution plots are generated to show why the prediction was made.

- **PDF Report Generation**

A complete clinical report with prediction and explanations is created.

Page 16

Medical Document Search Module Diagram



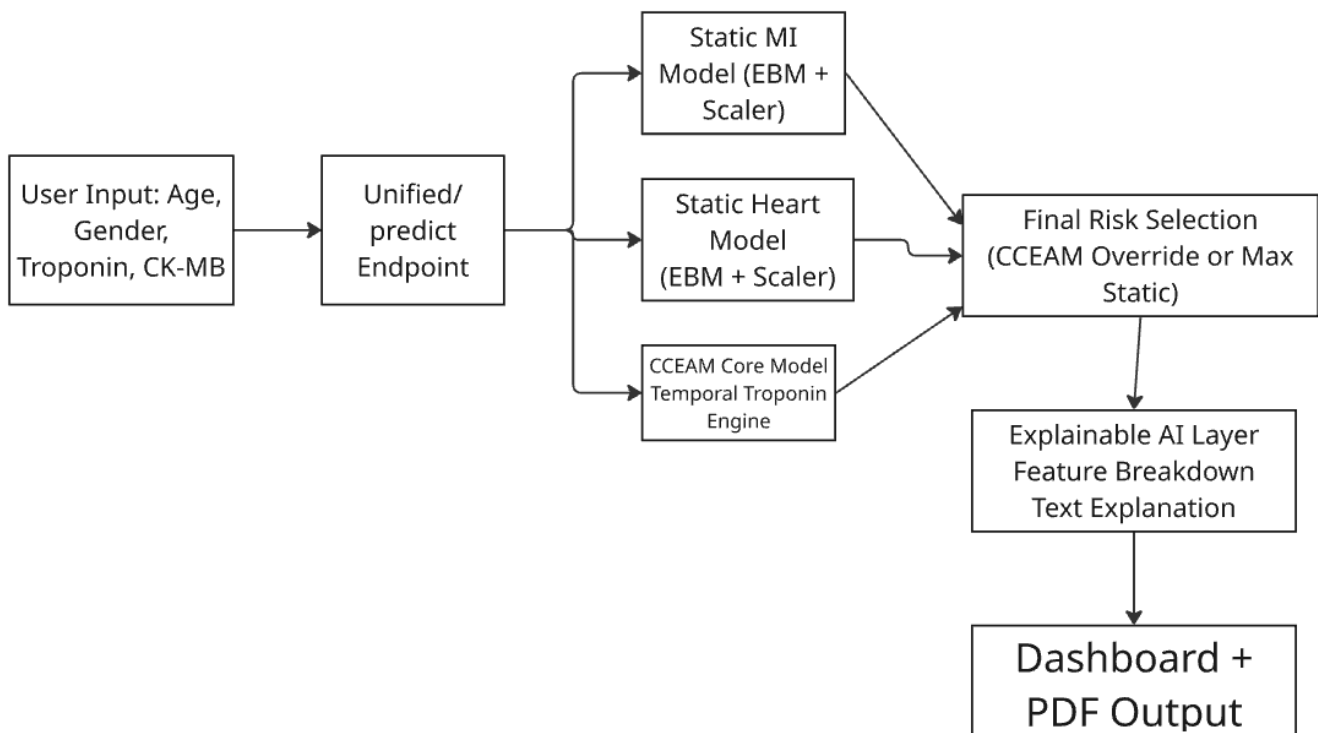
Page 17

Medical Document Search Module

- Developed a PubMed based medical document retrieval module using NCBI Entrez API.
- Extracts structured metadata including title, authors, journal name, publication year, and abstract.
- Implements semantic relevance ranking using Sentence-BERT embeddings.
- Uses cosine similarity to compute relevance between user query and research papers.
- Applies hybrid ranking strategy: latest publications prioritized, followed by relevance score.

Page 18

Risk Prediction Module Diagram



Page 19

Risk Prediction Module

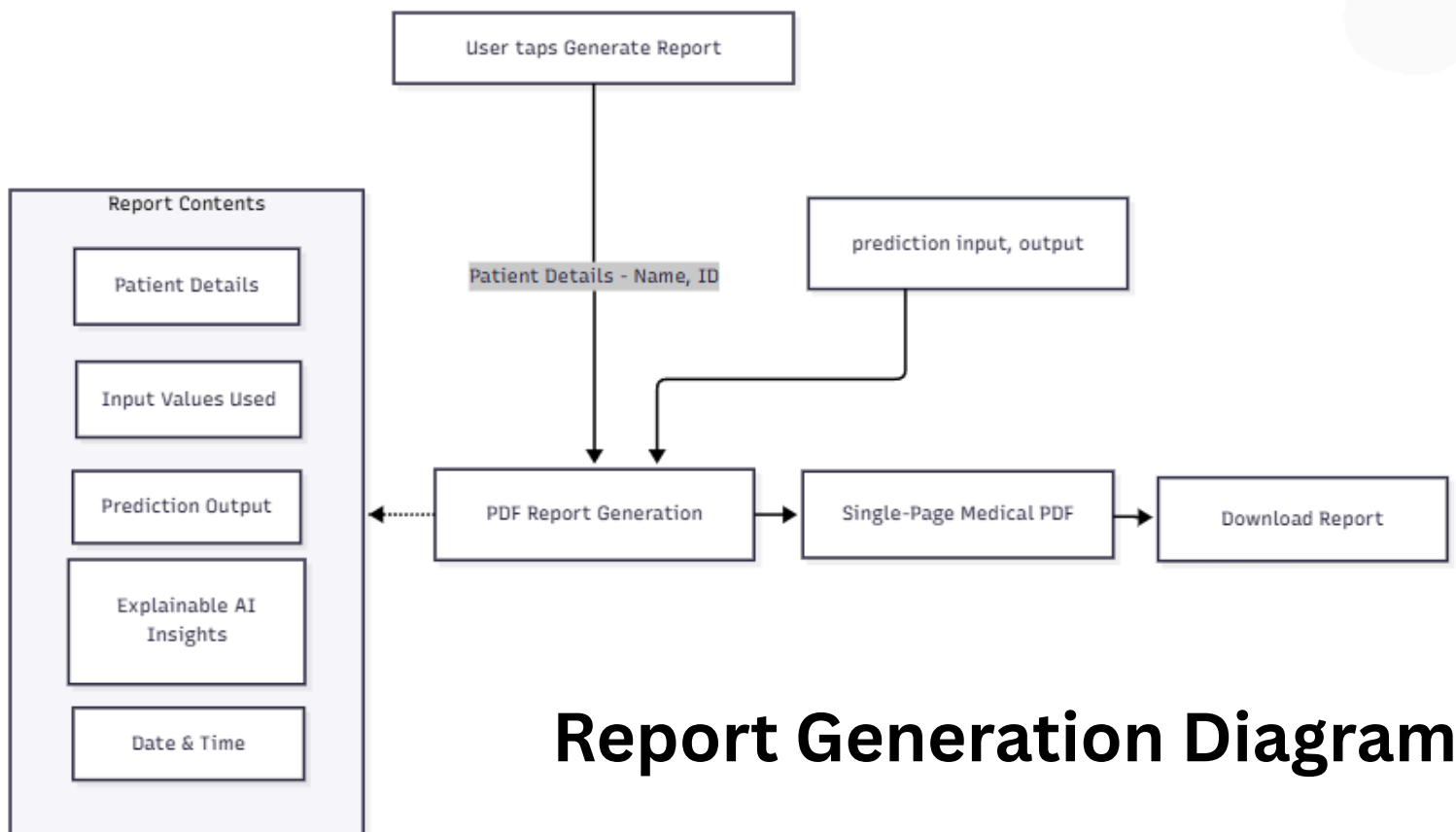
- Developed a unified cardiovascular risk prediction engine integrating static and temporal machine learning models.
- Accepts demographic inputs (age, gender) and clinical biomarkers including troponin, and CK-MB.
- Implements multi-path prediction logic using:
 - Static MI model (acute myocardial infarction detection)
 - Static heart risk model (chronic cardiovascular risk)
 - CCEAM temporal model (dynamic troponin progression analysis)
- Applies feature scaling pipelines before inference to ensure model consistency with training distribution.
- Utilizes Explainable Boosting Machine models to compute probability-based risk scores.
- Performs temporal troponin forecasting (1-hour and 2-hour predictions) and extracts dynamic features such as slope, delta, peak, and AUC for enhanced accuracy.

Page 20

Risk Prediction Module

- Applies probability fusion logic where CCEAM overrides static models when acute biomarkers are available; otherwise selects the maximum static risk.
- Generates explainable AI output including feature-level contributions and text-based clinical interpretation.
- Supports automated PDF report generation and literature recommendation based on computed risk profile.

Page 21



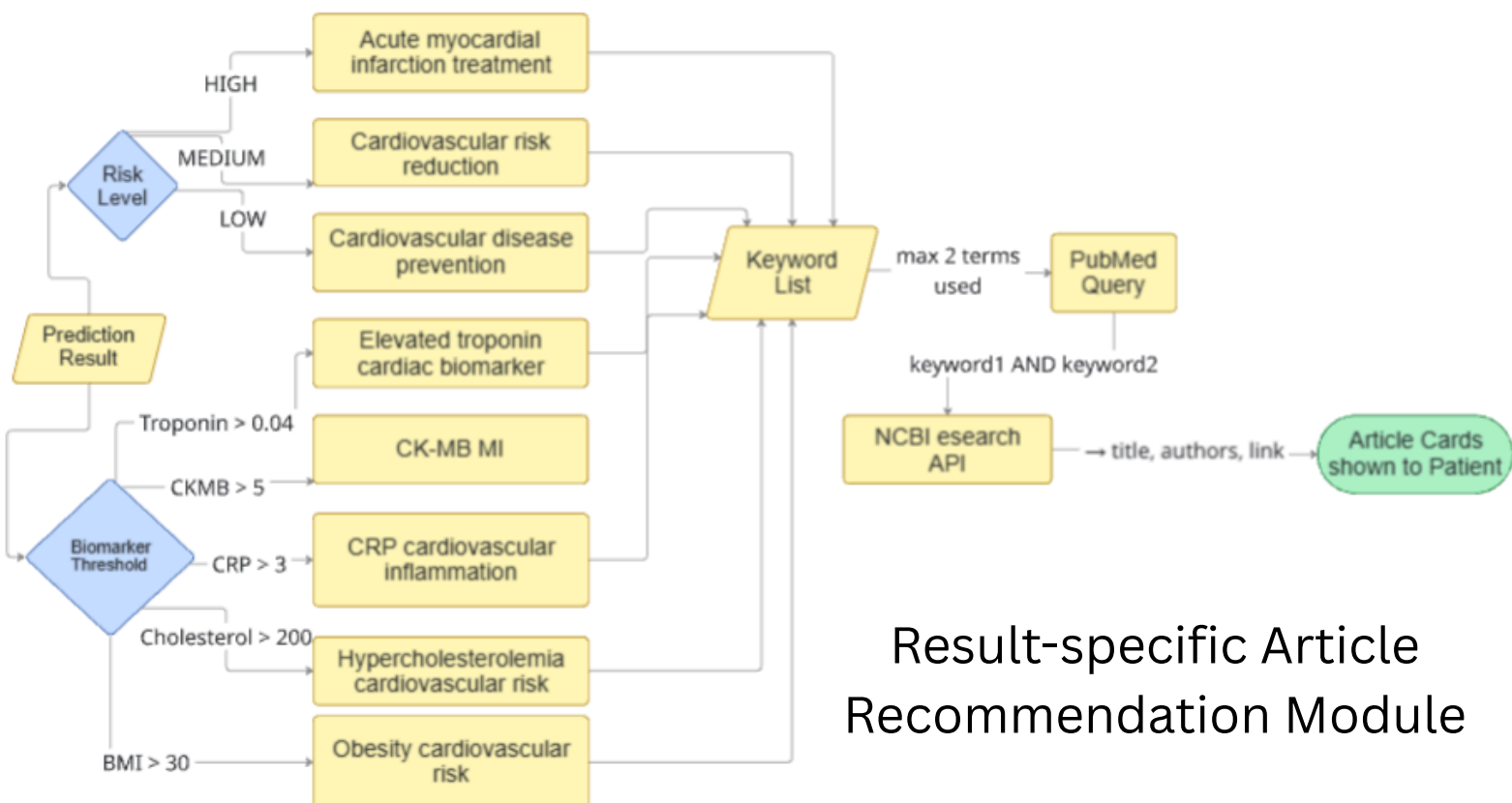
Report Generation Diagram

22

Report Generation Module

- Generates a structured, clinical-grade PDF report on demand, consolidating all prediction outputs into a single downloadable document.
- Reuses the exact final_prob computed during the prediction step — ensuring the PDF risk figure is always consistent with what is displayed on screen.
- Incorporates a troponin time-series (initial, 1-hour, and 2-hour projected values) extracted via temporal feature engineering, providing a dynamic view of cardiac biomarker progression.
- Translates model outputs into a human-readable text explanation using generate_text_explanation(), making the report interpretable without clinical expertise.
- Delivered directly to the browser as a binary file stream, enabling instant download without intermediate storage on the client side.

Page 23



Result-specific Article
Recommendation Module

Result-specific Article Recommendation Module

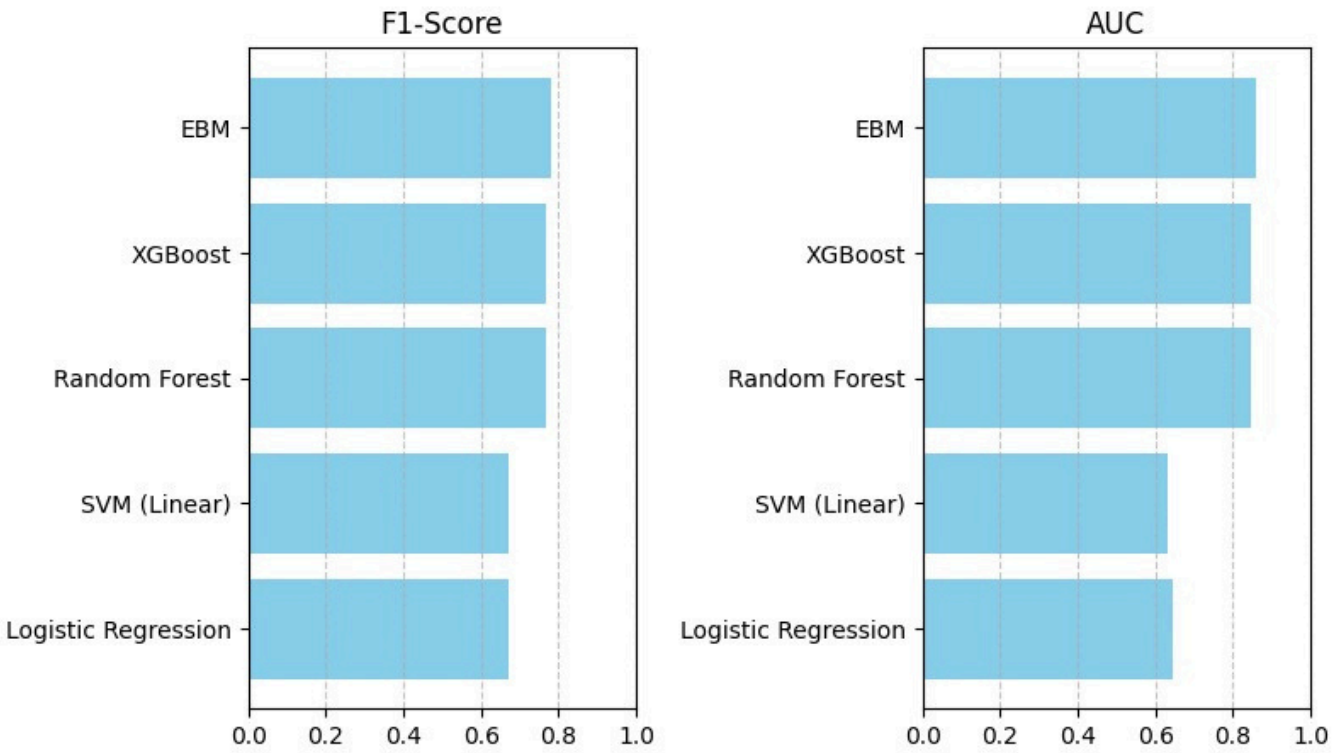
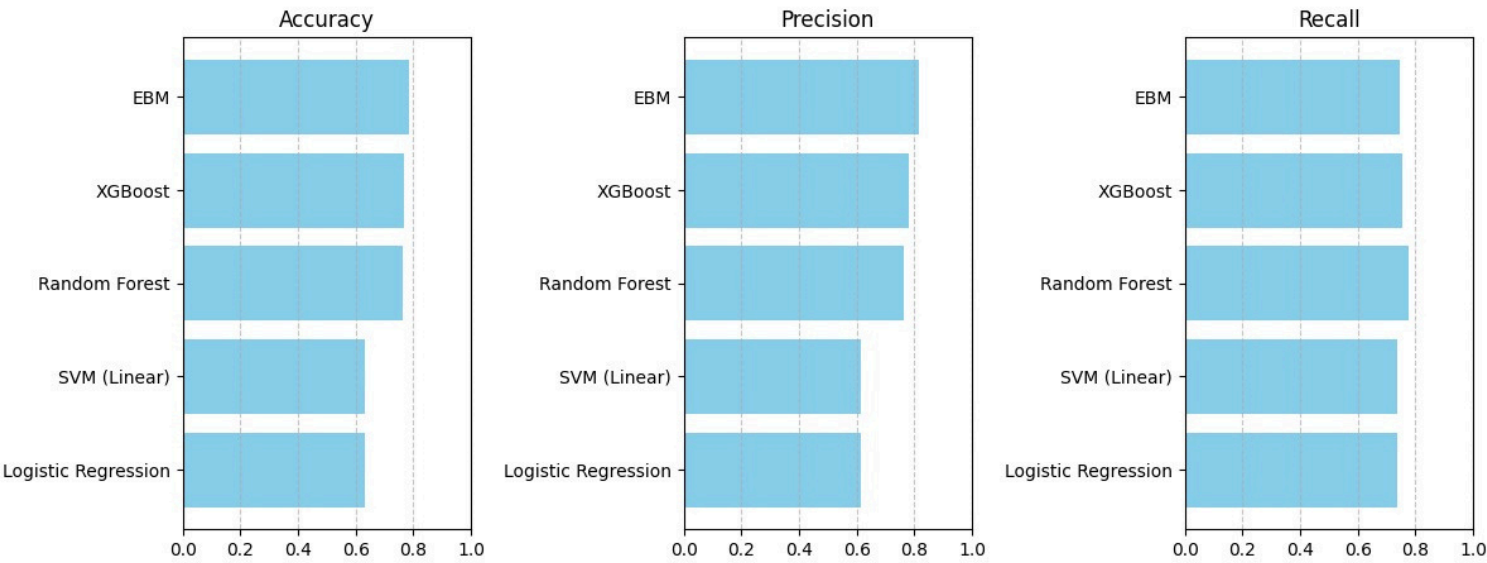
- Accompanies every recommendation with a plain-language "Why these articles?" explanation, contextualising the results to the patient's specific biomarker profile.
- Triggered automatically after every risk prediction — requiring no manual input from the patient.
- Dynamically extracts search keywords from the patient's risk category (HIGH / MEDIUM / LOW) and abnormal biomarker values (troponin, CK-MB).
- Constructs a focused PubMed query using up to the two most clinically relevant keyword phrases joined with an AND operator.
- Fetches and displays the top 5 most relevant research articles by querying the NCBI PubMed E-Utilities API (esearch + esummary endpoints).

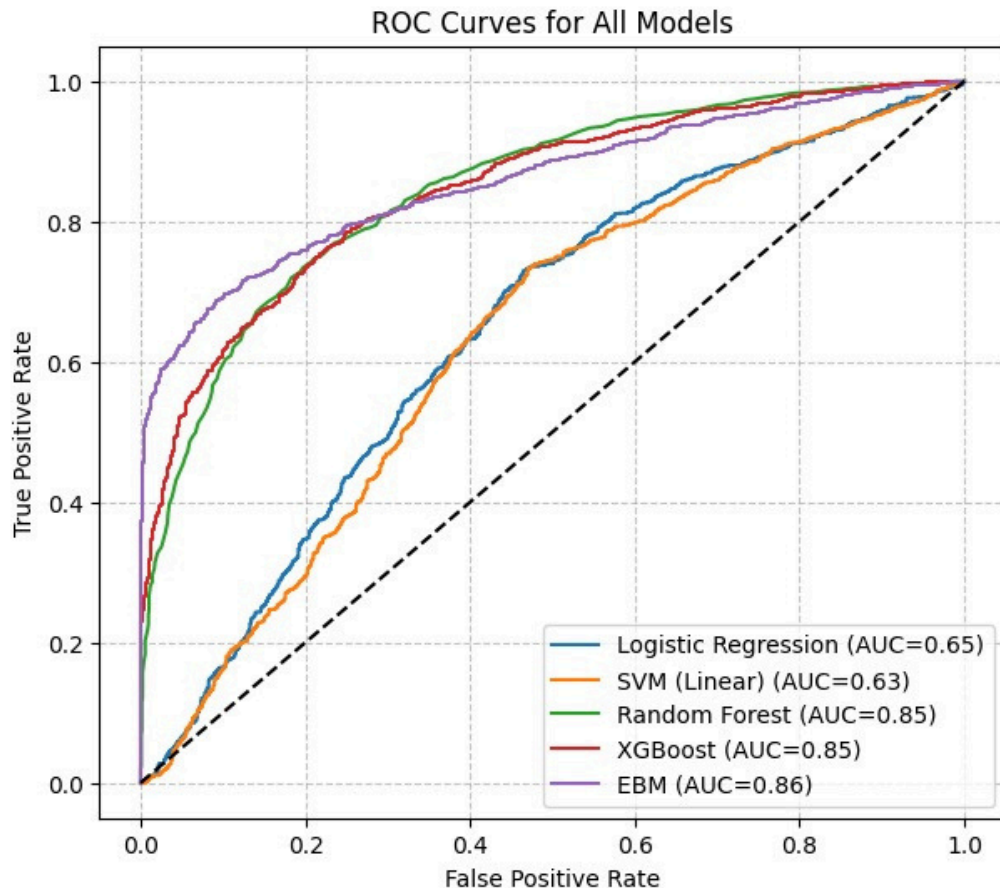
Page 25

RESULTS

Page 26

Model Performance Comparison





Page 29

--- CC-EAM Model Evaluation ---

Accuracy: 0.9597

Precision: 0.9623

Recall: 0.9559

F1-score: 0.9591

ROC-AUC: 0.9911

Page 30

OUTPUTS

Page 31

Reference Range of Biomarkers

Cardiac Troponin

Normal Level:

- < 0.04 ng/mL (for Troponin I in most labs)

Borderline:

- 0.04 – 0.39 ng/mL
- Possible cardiac injury (needs monitoring)

High (Suggestive of Myocardial Infarction):

- ≥ 0.40 ng/mL

CKMB

Normal Level:

- 0 – 5 ng/mL

Elevated:

- > 5 ng/mL

Strongly Suggestive of MI:

- Usually significantly elevated along with troponin rise

Page 32

TX

Welcome Back

Sign in to your dashboard

Patient

Clinician

Username

Enter your username

Password

Enter your password

Sign In

[Create a new account](#)

Troponix

127.0.0.1:5500/frontend/dashboard.html

Summarize

U

Lo

Dashboard

Cardiovascular Risk Assessment

Dashboard

Risk Assessment

CCEAM Troponin Risk

Nearby Hospitals

My Health Status

--%

Not Calculated

Medical Literature Search

Search PubMed research

Search medical research...

Search

Enter a medical term to search PubMed

My Health Overview

Your personalized health monitoring dashboard

Blood Biomarker ML Dashboard • Secure Health Analytics

Page 33

My Health Status

4.00%

LOW RISK

Medical Literature Search

Search PubMed research

Search medical research...

Search

Enter a medical term to search PubMed

Dashboard

Risk Assessment

CCEAM Troponin Risk

Nearby Hospitals

Cardiovascular Risk Assessment

Enter biomarker and patient details

Troponin (ng/mL)

0.02

CK-MB (U/L)

6

CRP (mg/L)

Total Cholesterol (mg/dL)

Triglycerides (mg/dL)

Homocysteine (μmol/L)

BMI (kg/m²)

Age

76

Gender

Male

Calculate Risk

Clear

Prediction: LOW RISK

Probability: 4.00%

Confidence: 1

Risk Summary

Your results indicate a low cardiovascular risk at this time. Most values are within acceptable ranges.

No individual biomarker is severely abnormal, but overall risk is based on combined factors.

Page 34

68

**Recommended Medical Literature**
Articles curated based on your risk assessment

Why these articles?

Beyond lipid lowering: Effects of PCSK9 inhibition on inflammation and HDL function.
Seidel M, Seibert FS, Doevelaar A
[View Article](#)

Genetic variants lowering lipoprotein(a) and downregulating interleukin-6 signalling are additively associated with a decreased risk of cardiovascular disease.
Xiang B, Zhang R, Zhou Y
[View Article](#)

Statin therapy in systemic lupus erythematosus: a meta-analysis of disease activity and inflammatory biomarkers.
Qian S, Wu J, Hu C
[View Article](#)

Coronary inflammation: Identification, clinical impact and possible therapeutical solutions.
Prati F, Biccirè FG, Mallus MT
[View Article](#)

Number of group psychotherapy sessions predicts inflammation changes in patients with coronary artery disease and comorbid depression.
Ronel J, Olliges E, Herrmann-Lingen C
[View Article](#)

DashboardRisk AssessmentCCEAM Troponin RiskNearby Hospitals

CCEAM Acute MI Risk
Temporal troponin prediction

Age49

GenderMale

Troponin 0h (ng/mL)0.1

CK-MB (U/L)23

Predict Risk

Download CCEAM Report (PDF)

Prediction: HIGH RISK
Probability: 99.00%
Troponin 1h: 0.14
Troponin 2h: 0.18

Myocardial Infarction (MI) Prediction Report

Patient Details

Age: 36.0
Gender: Male
Troponin: 0.08
CK-MB: 2.0

Prediction Output

Predicted MI Risk: 99.00% (HIGH)

Medical Interpretation

High probability of acute myocardial infarction (MI). Biomarkers strongly suggest myocardial injury. Immediate clinical evaluation is recommended.

Summary

The model predicts a high chance of heart attack. The results look serious and urgent medical attention is needed.

Explainable AI Breakdown (Feature Contributions)

Feature	Value	Contribution
Age	36.0	0.3015
Gender	1	0.8841
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Troponin	0.08	2.6308
Troponin_1h	0.10907	3.5867
Troponin_2h	0.14534	4.7795

Top Reasons

Predicted MI Risk: 99.00% (HIGH risk) Main factors increasing risk: - Troponin_2h (value=0.14534) increased risk - Troponin_1h (value=0.10907) increased risk - Troponin (value=0.08) increased risk

Reference Ranges

Troponin (0h, 1h, 2h): Normal < 0.04 ng/mL
CK-MB: Normal < 5 U/L

Nearby Heart Hospitals

Find cardiology hospitals near your location.

Find Hospitals Near Me

Ambalamugal Medical Aid Society (AMAS)

Address not available

Not available

3.31 km away

Open in Map

Urban Primary Health Centre, Chalikkavattom

Address not available

Not available

3.96 km away

Open in Map

Sunrise Hospital, Ernakulam

Address not available

Not available

4.15 km away

Open in Map

Nirmala Medical Mission Hospital

Address not available

Not available

4.43 km away

Medical Literature Search

Search PubMed research

normal range of troponin

Search

Cardiac Biomarker Kinetics in Patients after Coronary Artery Bypass Grafting with an Uncomplicated Course.

Denessen EJS, Swinnen B, Lamers SK, Delnoij TSR, van Doorn WPTM, Bekers O, Segers P, Maessen JG, van der Horst ICC, Heuts S, Mingels AMA

View Article

99th percentile upper reference limits of the 5th and 6th generation Elecsys® cardiac troponin T high-sensitivity assays in a single cohort of healthy blood donors.

Egger M, Schmid P, Kimbacher C, Black A, Gabriel C, Mueller T, Dieplinger B

View Article

Analytical performance of the Roche Elecsys TnT hs Gen 6 immunoassay.



ASSUMPTIONS

- Biomarker values are reliable and obtained from accurate lab tests.
- Training datasets are representative of real-world patient cases.
- The patient population used for training reflects similar demographic and clinical characteristics as the deployment population.
- The model is used as a decision-support tool and not as a standalone diagnostic replacement.
- External clinical factors not included in the dataset (e.g., ECG findings, imaging reports) are assumed not to drastically alter risk interpretation.

Page 39

WORK BREAKDOWN AND RESPONSIBILITIES

- **Literature Survey & Research – All Members**

Conducted background study on myocardial infarction prediction, biomarker-based diagnosis, explainable AI techniques, and existing clinical decision systems.

- **Dataset Collection & Preprocessing – Namitha**

Responsible for collecting biomarker datasets, cleaning missing values, handling class imbalance (SMOTE), feature selection, and preparing structured input data for model training.

- **Machine Learning Model Development & Evaluation – Nikitha**

Implemented ML algorithms (Random Forest, XGBoost, Logistic Regression,svm), performed training, hyperparameter tuning, cross-validation, and model performance comparison.

- **Medical Document Search Module – Namitha**

Designed and implemented the intelligent medical literature search module for retrieving relevant research papers and clinical references.

- **Frontend Development – Nandakishore**

Developed user interface for patient data input, prediction visualization, and interactive dashboard display.

Page 40

WORK BREAKDOWN AND RESPONSIBILITIES

- **Hospital Finder Module – Nandakishore**

Implemented hospital search functionality for identifying nearby cardiac care centers.

- **CCEAM Development - Nikitha**

Developed an additive model for heart attack risk detection

- **Report Generation Module – Smiya**

Developed automated medical report generation based on prediction results and biomarker analysis.

- **Hardware Research – All Members**

Explored feasibility of integrating the system with a portable biomarker testing device for real-time heart attack risk detection.

- **Prediction-Based Article Recommendation Module - Smiya**

Developed an automated article recommendation module based on the user's prediction

- **System Integration – Nandakishore**

Integrated backend ML models, XAI components, UI and additional modules into a unified system.

- **Research Paper & Documentation – All Members**

Prepared technical documentation, maintained reports, and drafted research paper for publication^{Page 41}



HARDWARE REQUIREMENTS

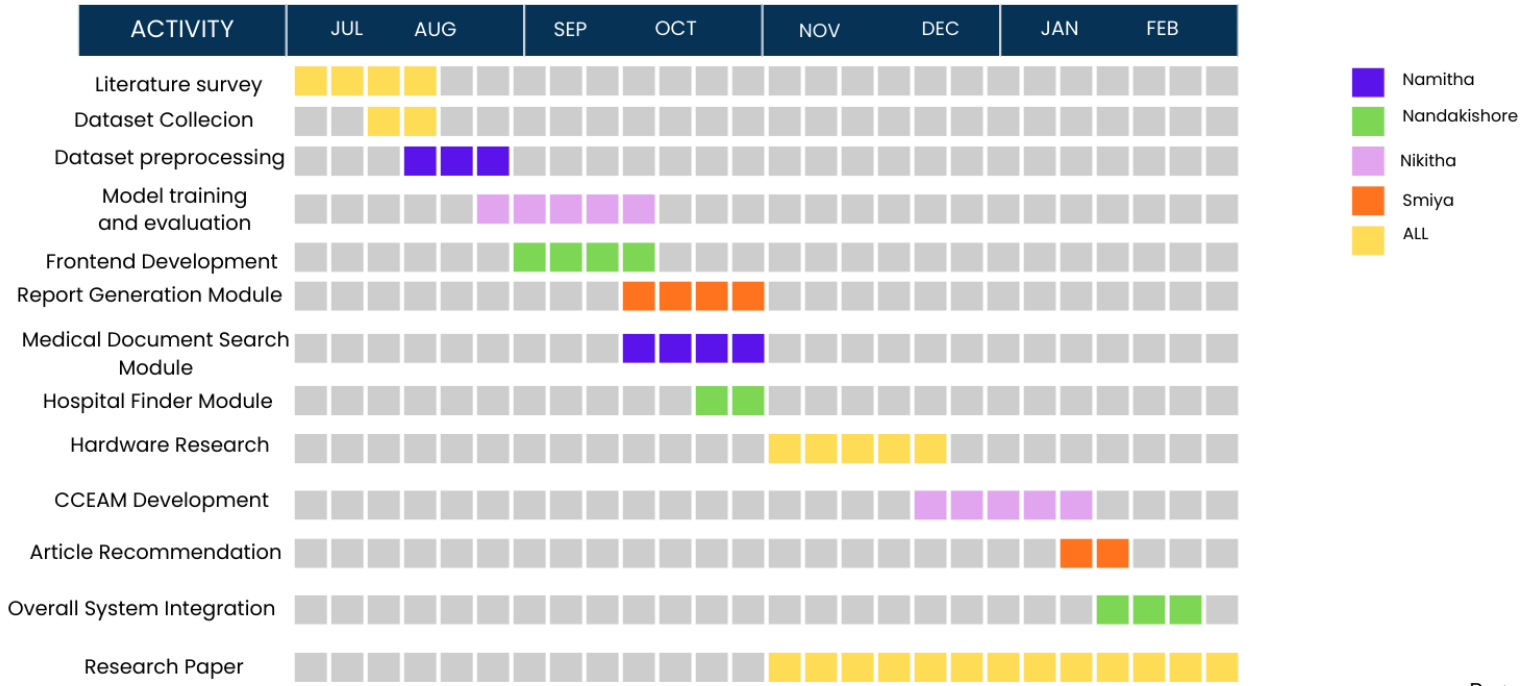
- Intel i5 / i7 processor or equivalent
- Minimum 8 GB RAM (16 GB recommended for model training)
- 256 GB storage or higher
- Standard input/output devices (keyboard, mouse, monitor)
- Internet connectivity for dataset access and deployment



SOFTWARE REQUIREMENTS

- Operating System: Windows / Linux
- Programming Language: Python, JavaScript, HTML, CSS
- Frameworks & Libraries: FastAPI, Node.js (Express.js), Scikit-learn, NumPy
- Development Environment: VS Code, Node.js
- Web Framework (for interface): HTML5, CSS3, Vanilla JavaScript
- Database: SQL (for authentication)

Gantt Chart





RISKS AND CHALLENGES

- Variations in biomarker thresholds across different populations, age groups, and comorbid conditions.
- Integration with real-time hospital systems may require additional clinical validation and regulatory approval.
- Risk of data imbalance (fewer MI cases compared to non-MI cases) affecting model performance.
- Data privacy and security concerns while handling sensitive patient health information.
- Potential dataset bias leading to reduced generalization across diverse demographic groups.
- Clinical acceptance challenges due to skepticism toward AI-based decision systems.
- Model performance degradation due to data drift in real-world deployment.
- Incorrect risk threshold selection may either increase false positives or false negatives.

Page 45



FUTURE SCOPE

- Development of a portable hardware device for real-time blood biomarker analysis.
- Expansion to multi-modal prediction using biomarkers + ECG + clinical history.
- Large-scale clinical validation across diverse populations.
- Integration with hospital systems as a Clinical Decision Support Tool.

Page 46



CONCLUSION

- This project presents an AI based heart attack prediction and detection system powered by explainable ai.
- The app analyzes the received biomarker data using CC-EAM model developed by us and provides users with clear, explainable outputs, helping them understand whether immediate medical attention is required.
- Additional app features include a medical search engine that answers user queries using verified scientific journals and a location-based service to identify the nearest hospital.
- Overall, the system demonstrates an interpretable, and user-centric decision-support approach aimed at enabling timely action during potential cardiac events.

Page 47

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Page 48

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Page 49



Thankyou

Page 50

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.
2. **PO2: Problem Analysis:** Identify, formulate, review research literature and ana-

lyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.
4. **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.
5. **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.
6. **PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for their impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.
7. **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.
8. **PO8: Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9. **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language and learning differences.
10. **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.
11. **PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES

S.No	Description	Bloom's Taxonomy Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND PROGRAM SPECIFIC OUTCOMES (PSOs)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	3	2		2	3	2	1	2	1	1	2	2	
CO3							3	2	2	3				3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	2	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	Level	Justification
101003/ CS822U.CO1- PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1- PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1- PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1- PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1- PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1- PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1- PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1- PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1- PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1- PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1- PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1- PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1- PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.

Mapping	Level	Justification
101003/ CS822U.CO2- PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2- PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2- PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2- PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2- PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2- PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2- PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2- PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2- PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2- PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2- PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2- PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3- PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3- PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3- PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3- PO11	Low	Team interaction supports learning from peers and adapting new approaches.

Mapping	Level	Justification
101003/ CS822U.CO3- PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4- PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4- PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4- PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4- PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4- PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4- PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4- PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4- PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4- PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4- PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4- PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5- PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5- PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5- PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5- PO4	Low	May involve limited investigation or validation of proposed ideas.

Mapping	Level	Justification
101003/ CS822U.CO5- PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5- PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5- PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5- PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6- PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6- PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6- PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6- PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6- PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6- PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6- PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6- PSO3	High	Communication of innovative computer science solutions and research outcomes.